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Park et al.

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(54) **ELECTRONIC DEVICE INCLUDING
BRACKET FORMED OF METAL MATERIAL**

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H01Q 1/24 (2006.01)

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(2013.01); **G06F 1/203** (2013.01); **H01Q**
1/243 (2013.01)

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(Continued)

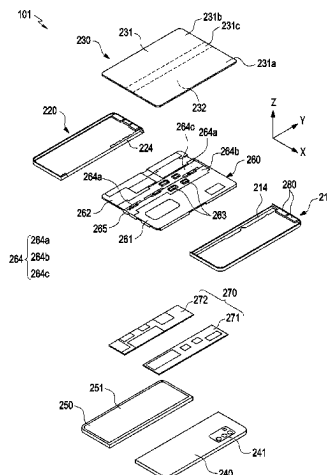
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(57) **ABSTRACT**

An electronic device is provided. The electronic device includes a foldable housing including a hinge structure, a first housing connected to the hinge structure and including a first side wall structure and a first bracket located inside the first side wall structure, and a second housing connected to the hinge structure and including a second side wall structure and a second bracket located inside the second side wall structure, the second housing being folded with the first housing about the hinge structure, a display panel extending from the first housing to the second housing across the hinge structure and configured to output a screen, and a second antenna structure including a second ground and a second feeding portion disposed on the second side wall structure of the second housing, wherein the first side wall structure of the first housing and the second side wall structure of the second housing may include a first metal material, and the first bracket of the first housing includes the first metal material, and the second bracket of the second housing may include a second metal material.

18 Claims, 14 Drawing Sheets



(58) **Field of Classification Search**

CPC G06F 1/1698; H01Q 1/243; H01Q 1/24;
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H04M 1/026; H04M 1/0268; H04M
1/0277; H05K 7/20; H05K 5/04; H05K
9/00; H05K 7/20336; H05K 7/20409;
H05K 9/002

See application file for complete search history.

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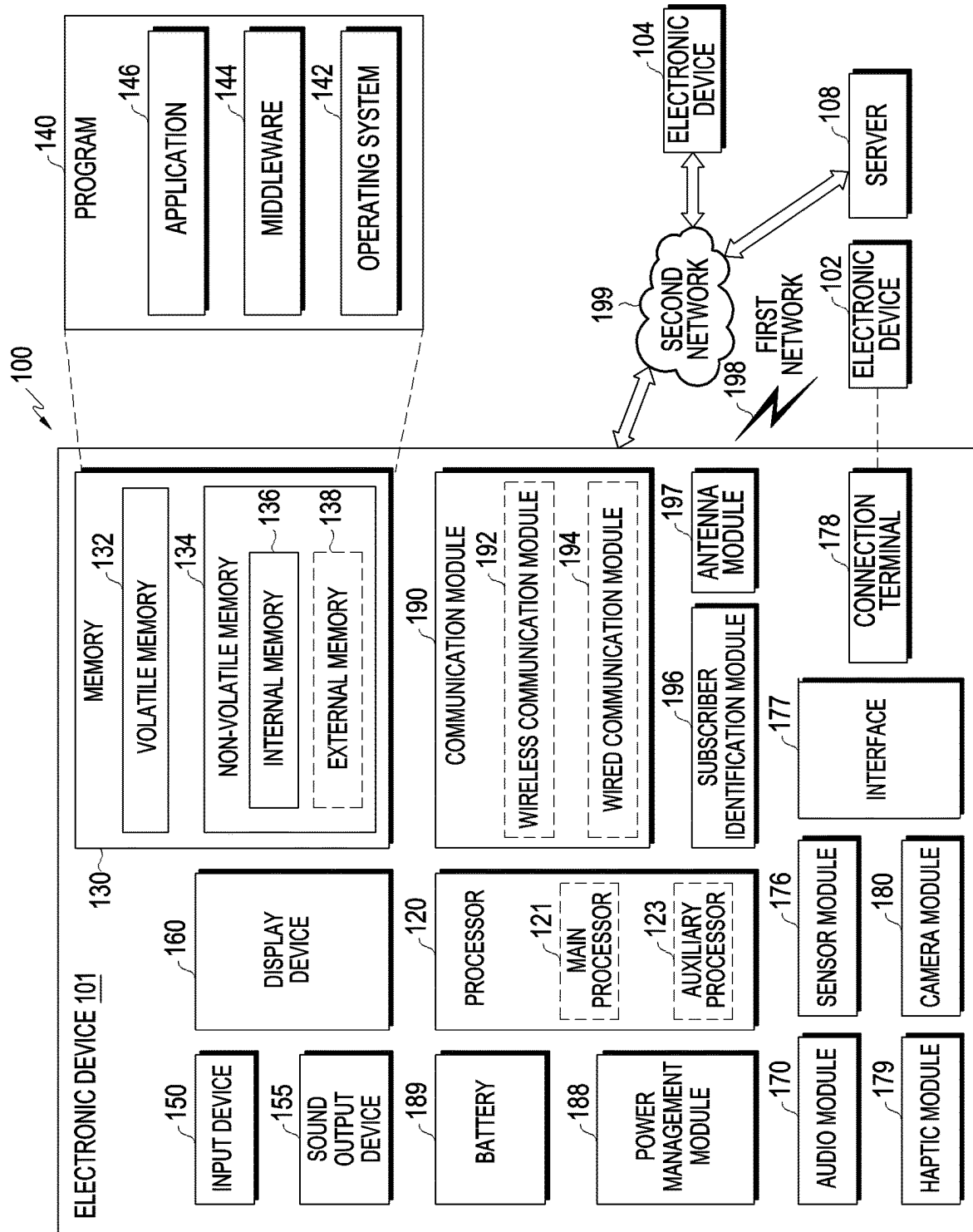


FIG. 1

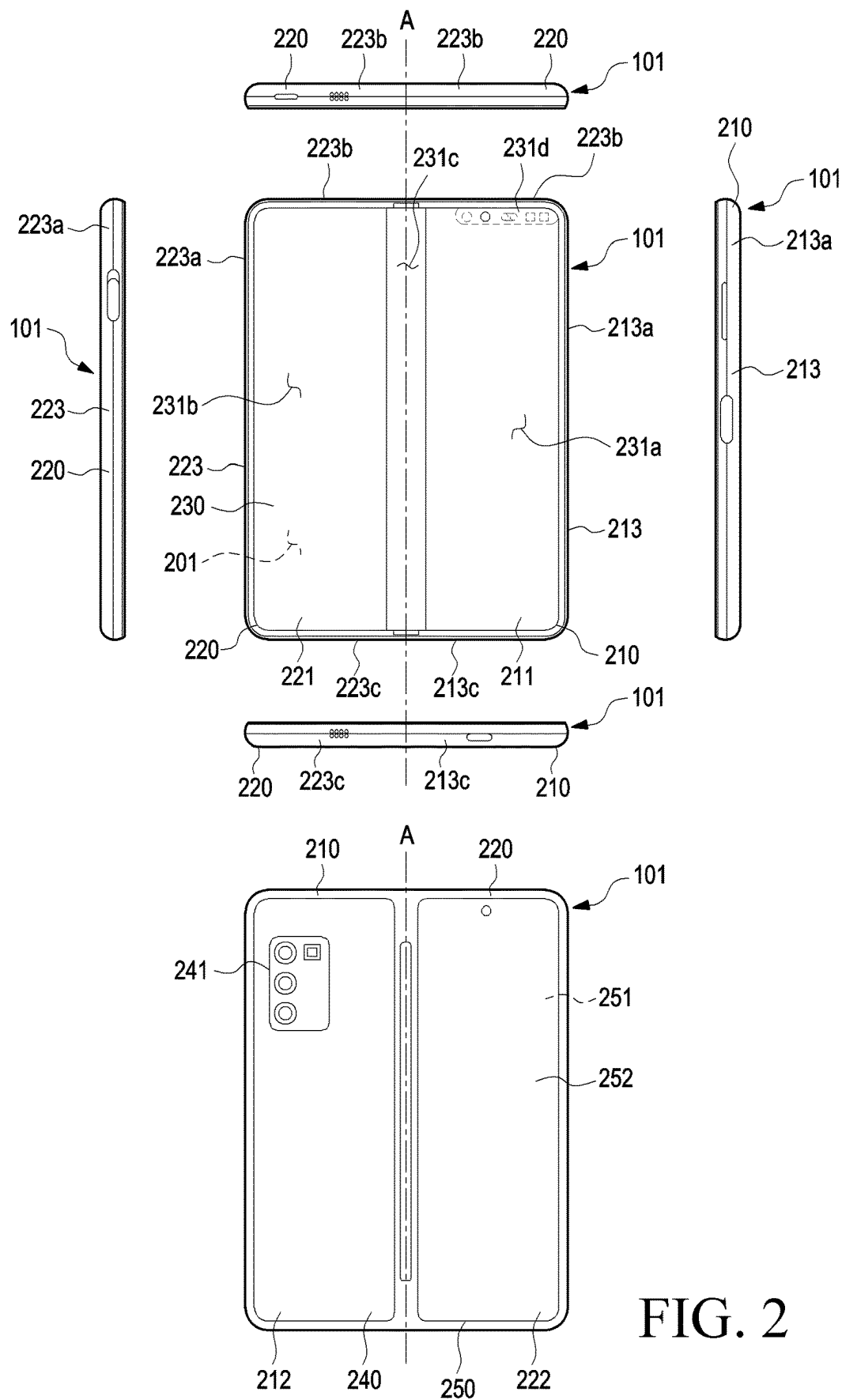


FIG. 3

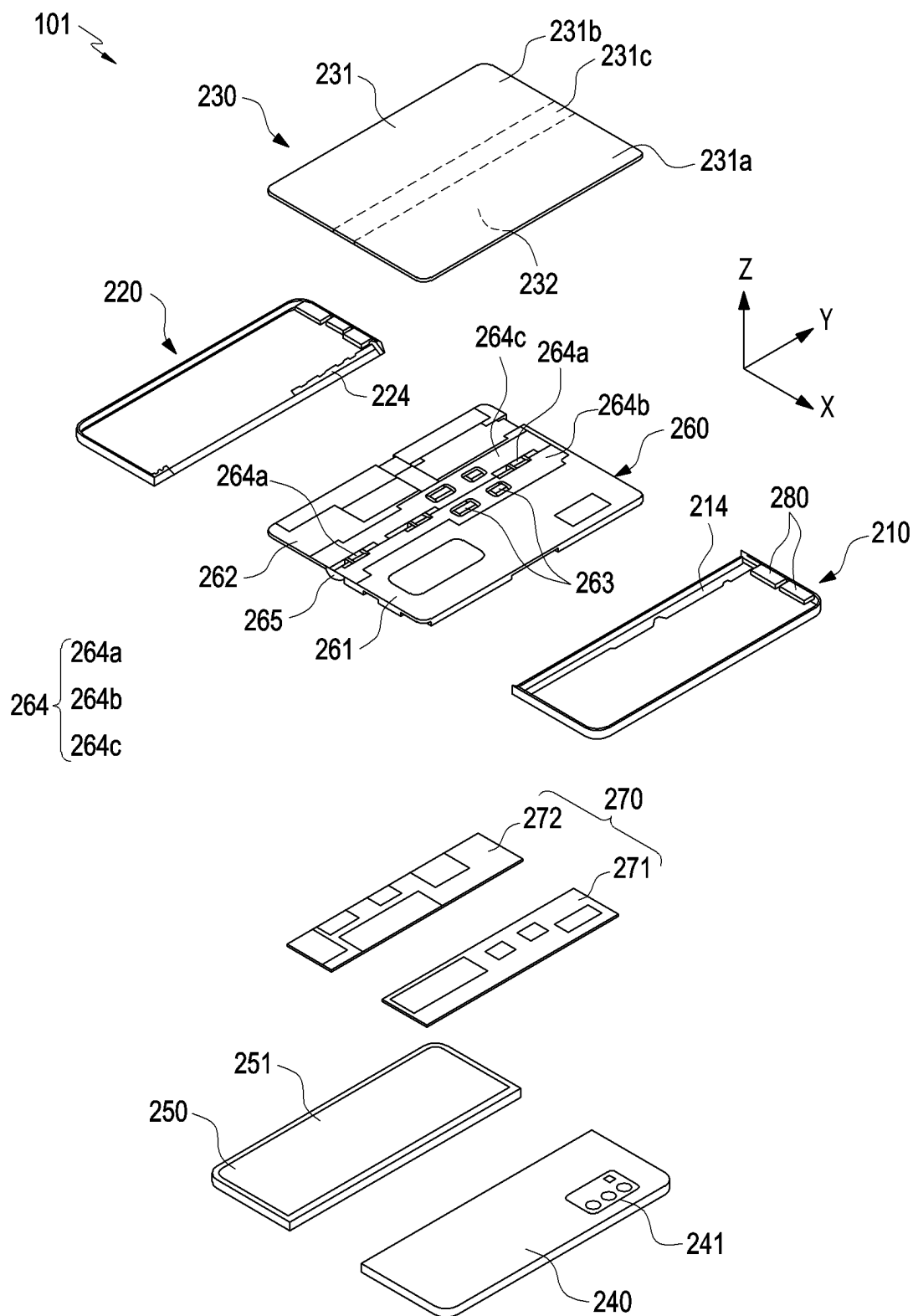


FIG. 4

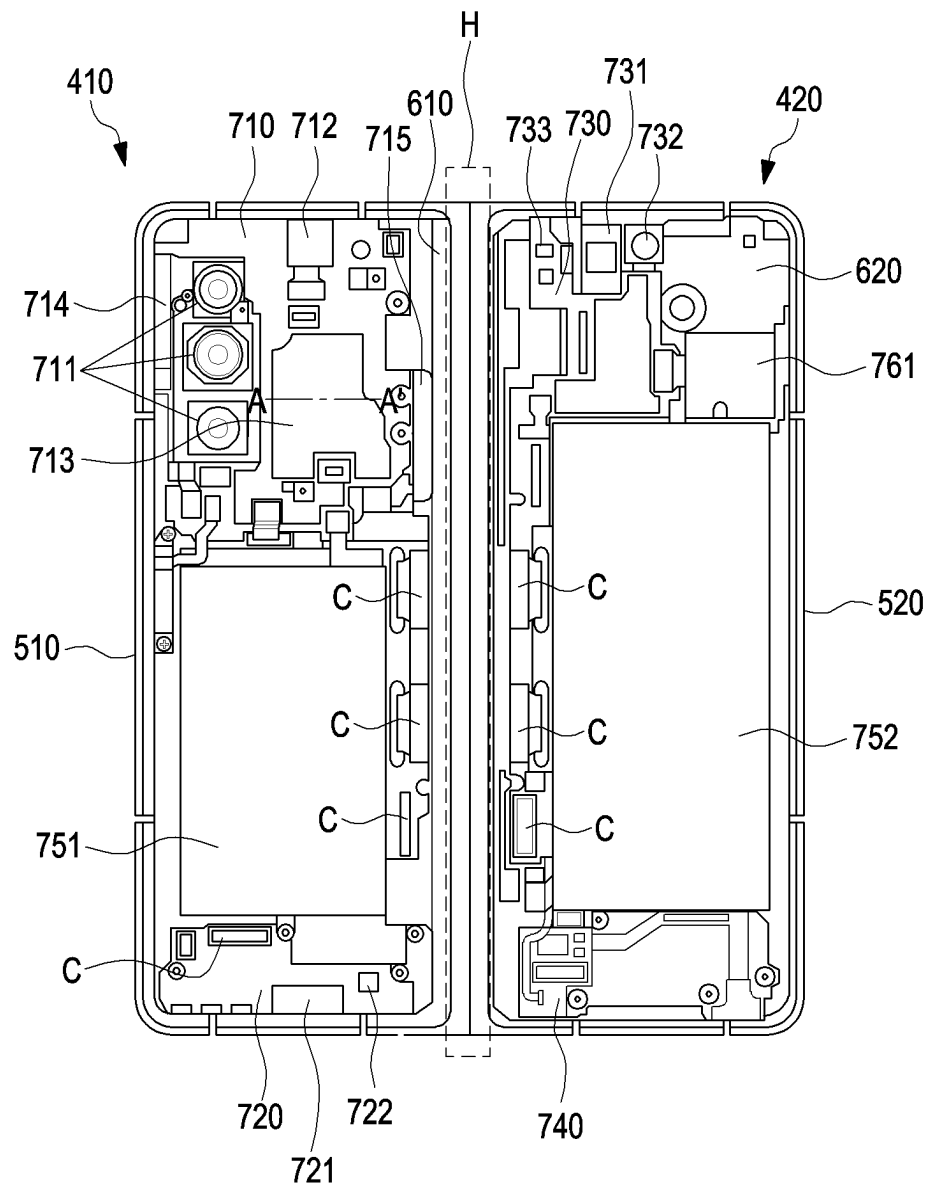


FIG. 5

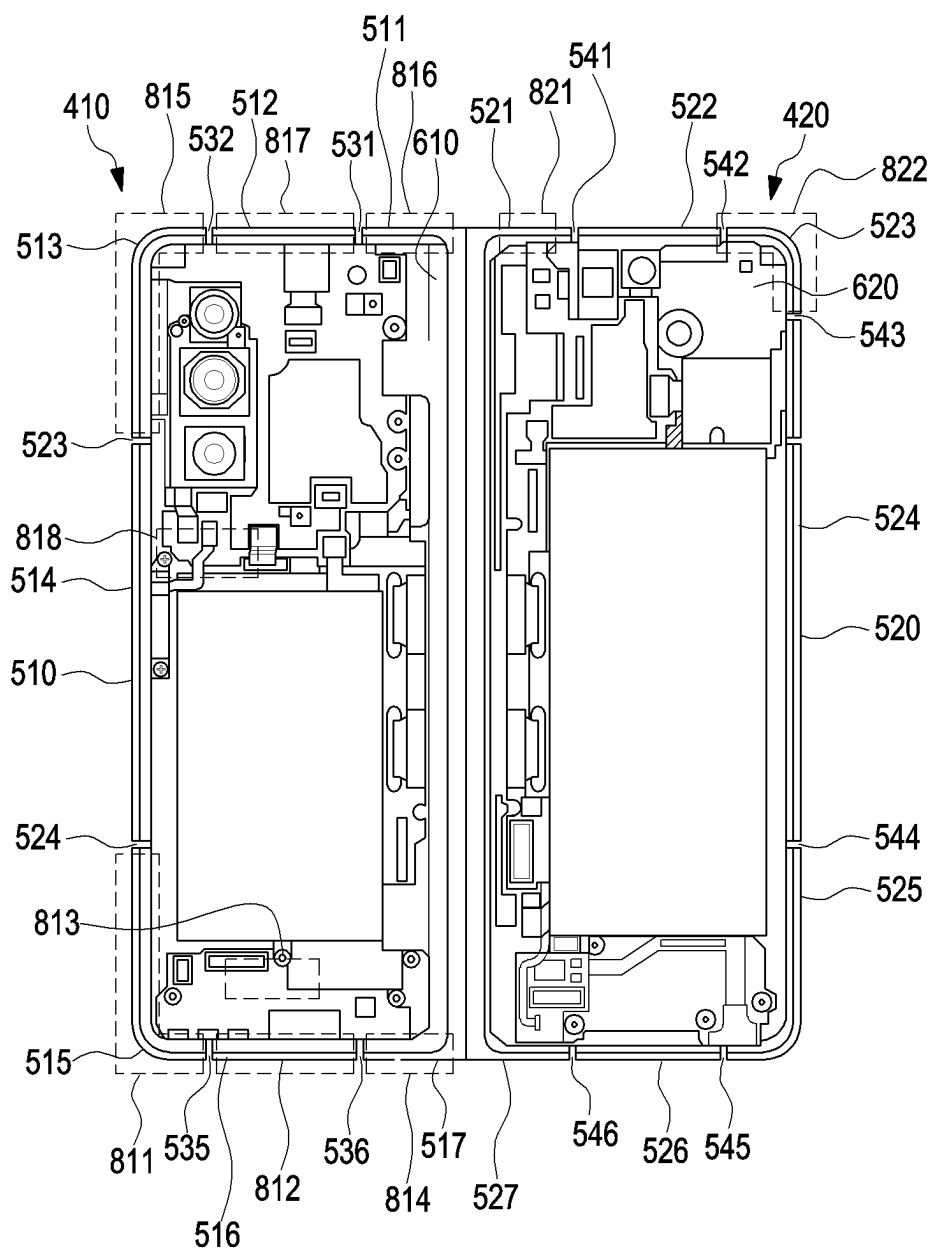


FIG. 6

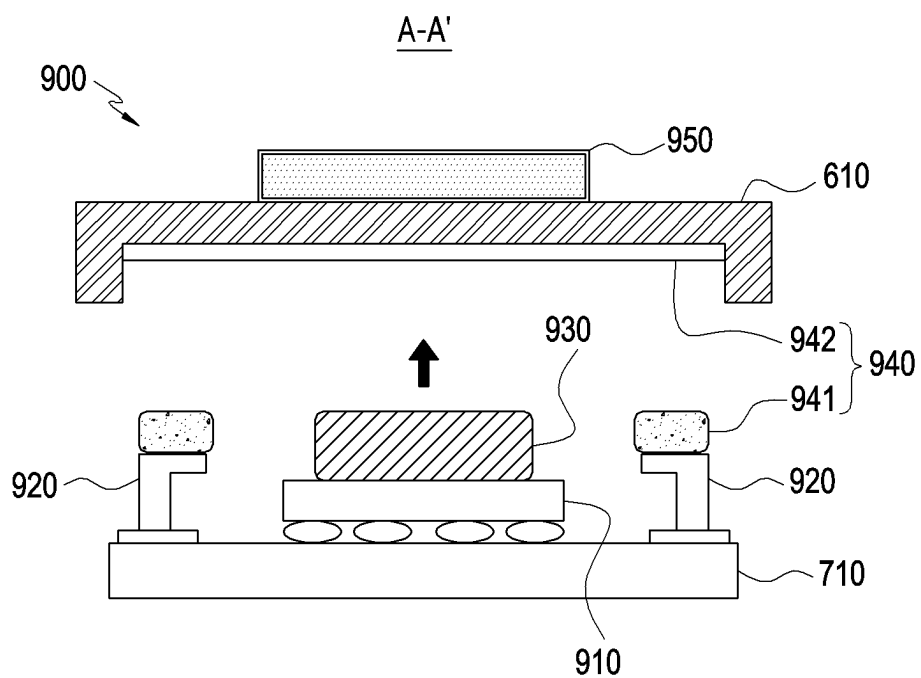


FIG. 7

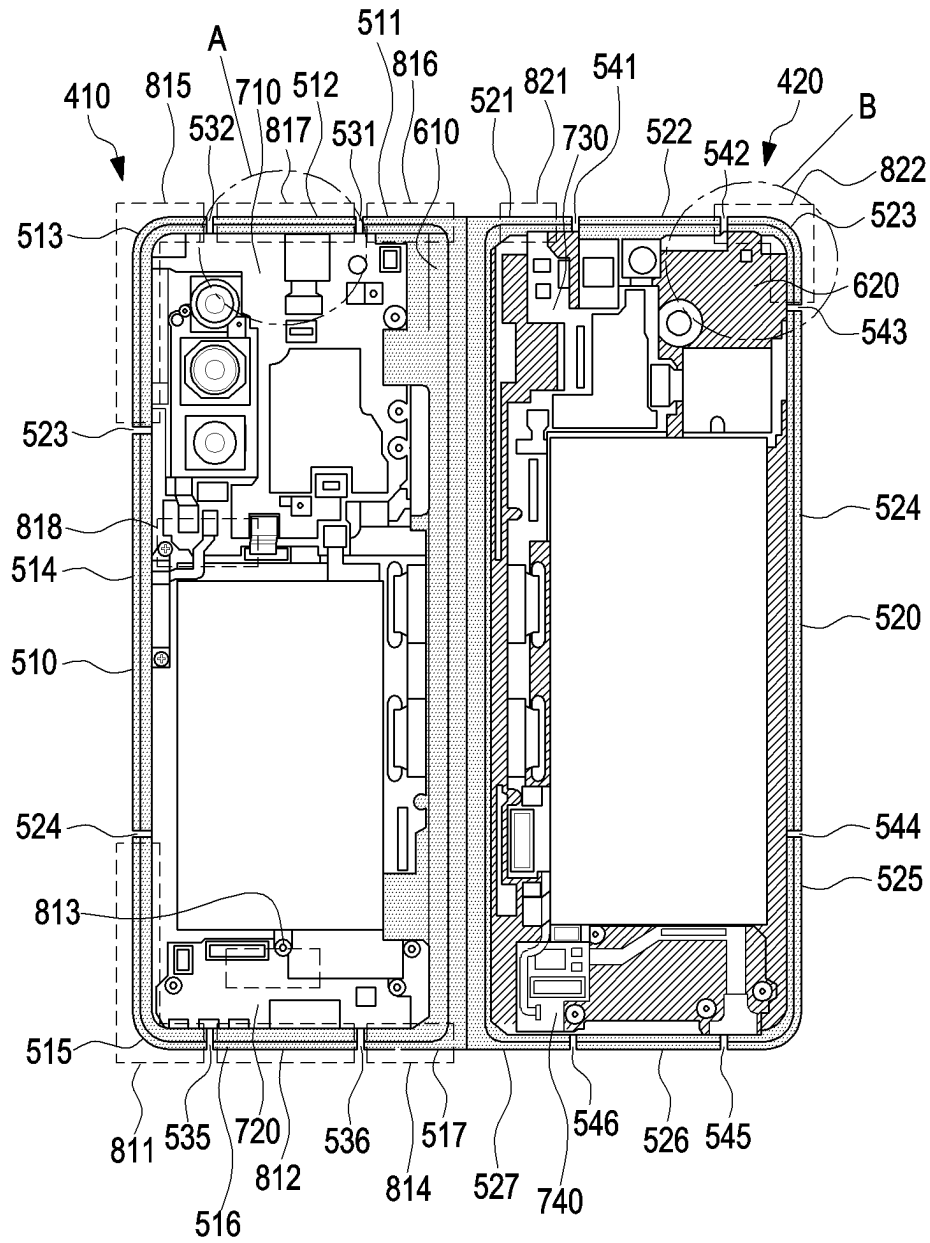


FIG. 8

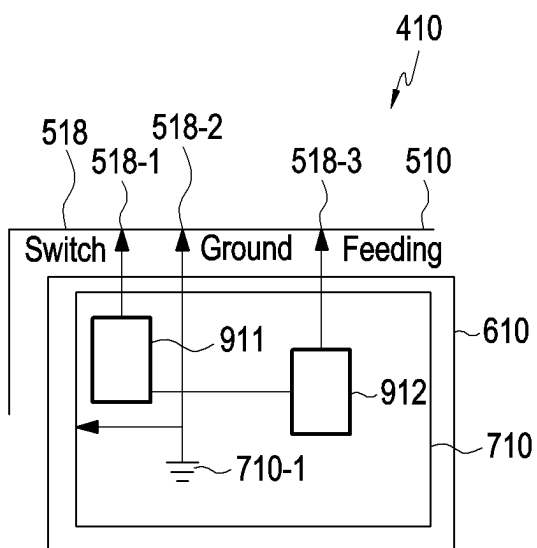


FIG. 9A

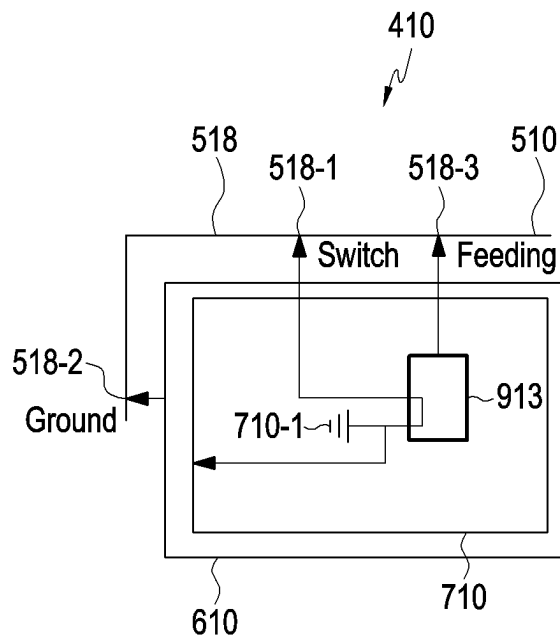


FIG. 9B

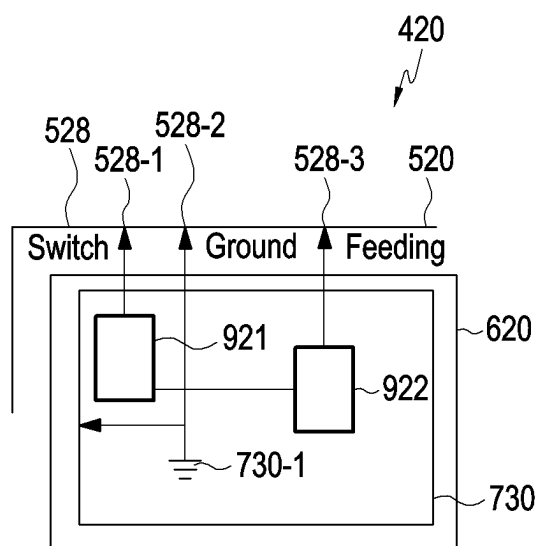


FIG. 10A

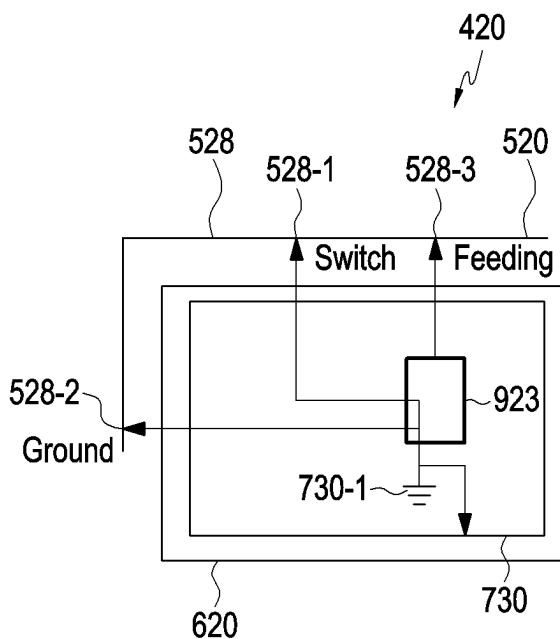


FIG. 10B

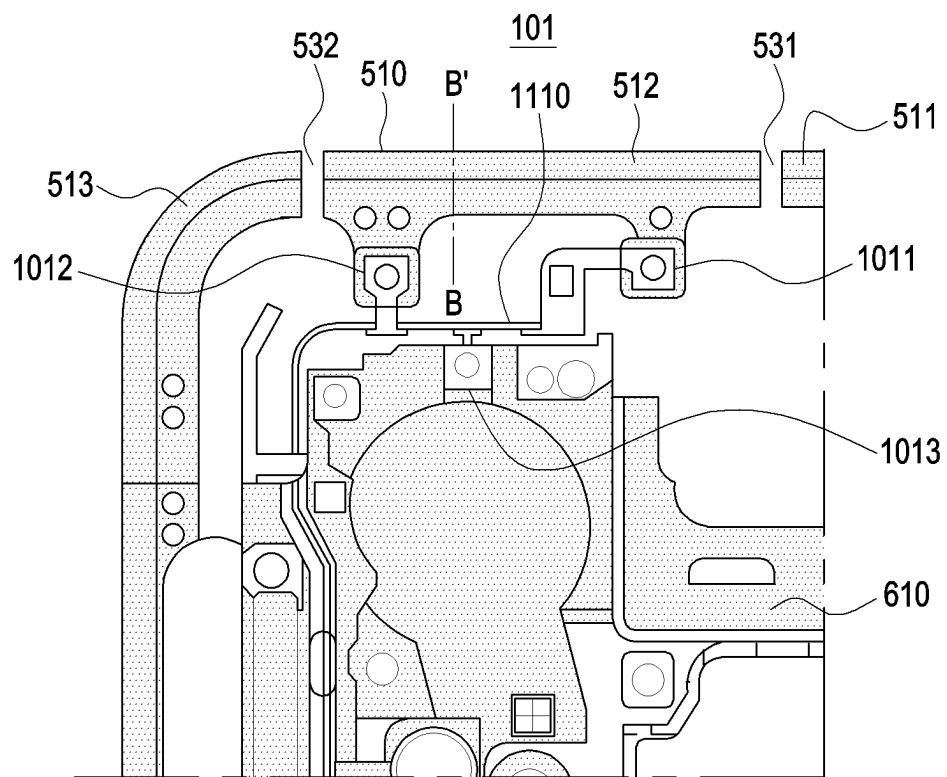


FIG. 11

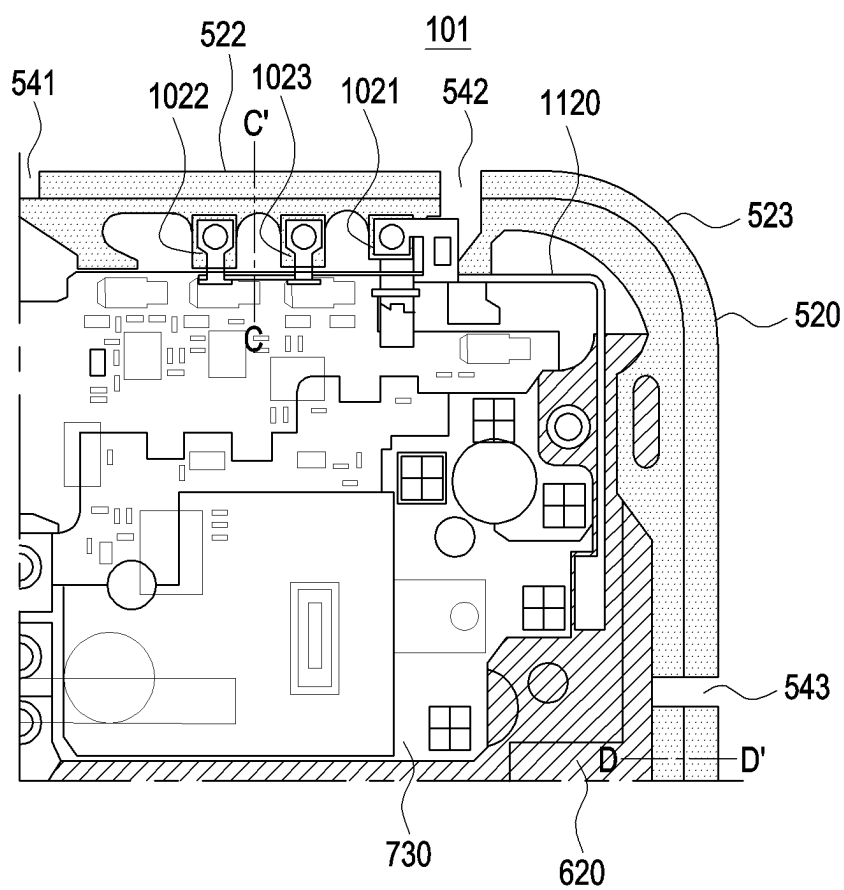


FIG. 12

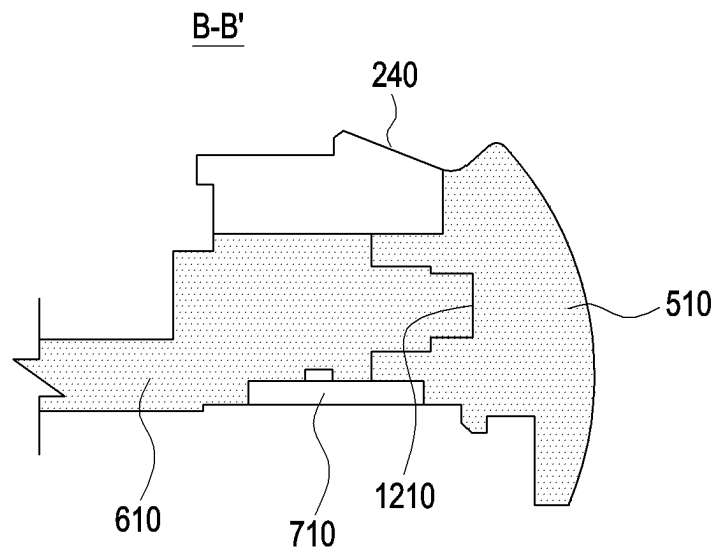


FIG. 13

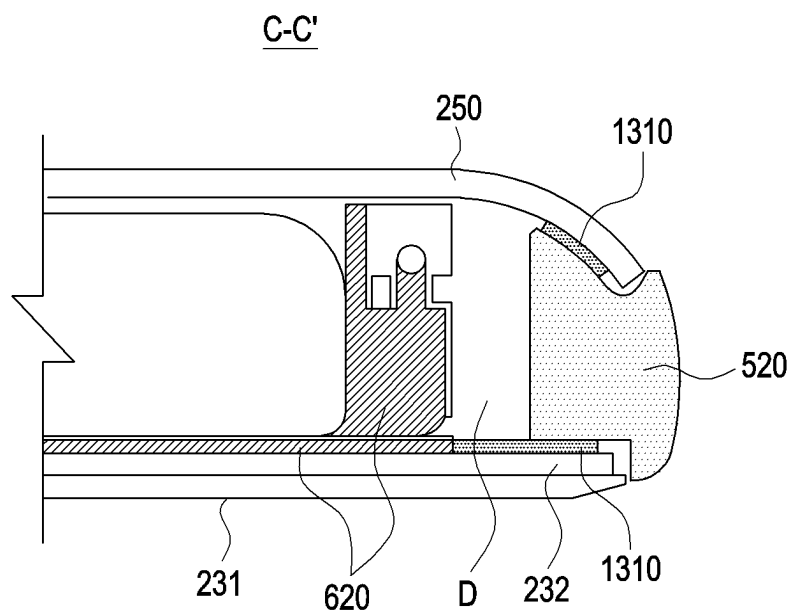


FIG. 14

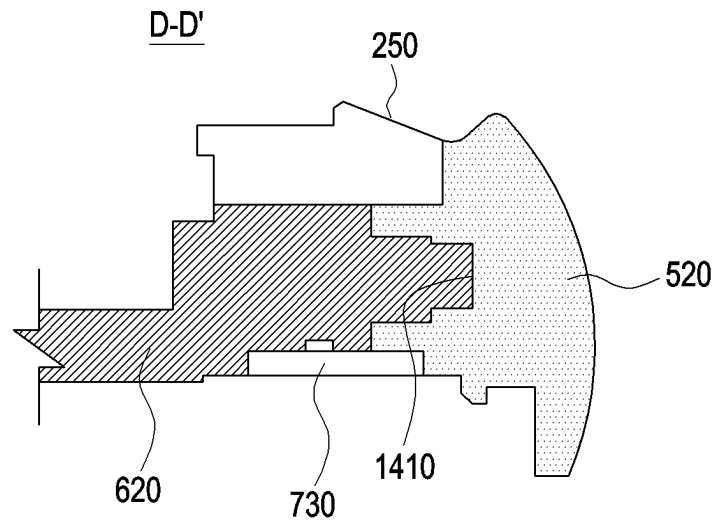


FIG. 15

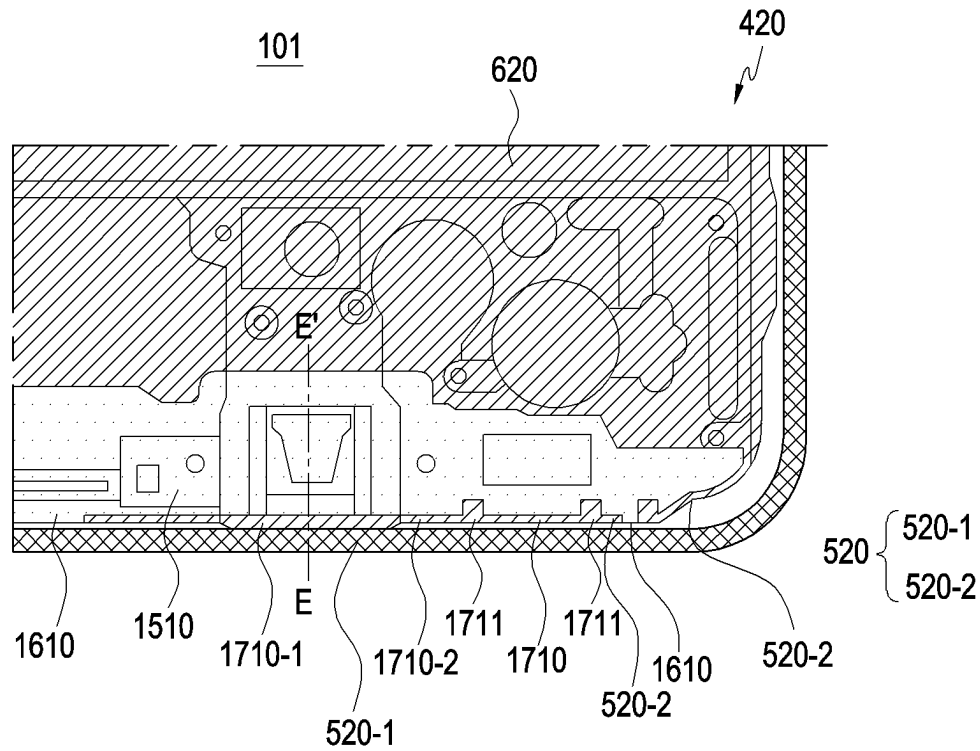


FIG. 16

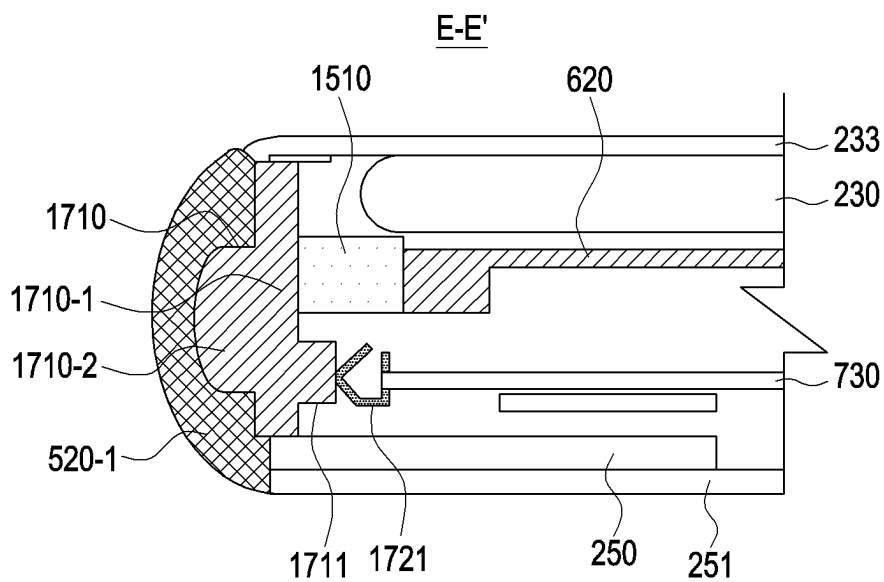


FIG. 17

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**ELECTRONIC DEVICE INCLUDING
BRACKET FORMED OF METAL MATERIAL****CROSS-REFERENCE TO RELATED
APPLICATION(S)**

This application is a continuation application, claiming priority under § 365(c), of an International application No. PCT/KR2022/012344, filed on Aug. 18, 2022, which is based on and claims the benefit of a Korean patent application number 10-2021-0164954, filed on Nov. 25, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

The disclosure relates to an electronic device including a bracket formed of a metal material. More particularly, the disclosure relates to an electronic device including brackets formed of different metal materials according to structural characteristics of the electronic device.

BACKGROUND ART

Due to the remarkable development of information communication technology, semiconductor technology, and the like, the distribution and use of various electronic devices are rapidly increasing. More particularly, recent electronic devices are being developed such that users are capable of communicating with each other while carrying the electronic devices.

Typically, an electronic device may mean a device that performs a specific function according to a program provided therein (e.g., an electronic scheduler, a portable multimedia reproducer, a mobile communication terminal, a tablet PC, an image/sound device, a desktop/laptop PC, or a vehicle navigation system), as well as a home appliance. The above-mentioned electronic devices may output, for example, information stored therein as sound or an image. As the degree of integration of electronic devices has increased and ultra-high-speed and large-capacity wireless communication has become popular, multiple functions have recently come to be provided in a single electronic device, such as a mobile communication terminal. For example, various functions, such as an entertainment function (e.g., a game function), a multimedia function (e.g., a music/video reproducing function), a communication and security function (e.g., mobile banking), a schedule management function, and an e-wallet function, are integrated in a single electronic device, in addition to a communication function. Such an electronic device is being miniaturized so that a user can conveniently carry the electronic device.

Recently, a portable electronic device, such as a smartphone may include housing structures having different structures based on a hinge structure, including a flexible display. Accordingly, measures for satisfying conditions required according to the structural characteristics of each housing structure of the electronic device may be considered.

The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is

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made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

**DETAILED DESCRIPTION OF THE
INVENTION****Technical Problem**

An electronic device including a flexible display may include housing structures having different structures based on a hinge structure. For example, one housing structure includes a printed circuit board on which a plurality of heating elements are stacked, so the thermal conductivity of a metal material constituting a bracket that supports the printed circuit board may be important, and the other housing structure includes a sub-display, so the specific gravity of a metal material constituting a bracket that supports the sub-display may be important.

Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device including brackets formed of different metal materials according to structural characteristics of the electronic device.

Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

Technical Solution

In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a foldable housing including a hinge structure, a first housing connected to the hinge structure and including a first side wall structure and a first bracket located inside the first side wall structure, and a second housing connected to the hinge structure and including a second side wall structure and a second bracket located inside the second side wall structure, the second housing being folded with the first housing about the hinge structure, a display panel extending from the first housing to the second housing across the hinge structure and configured to output a screen, and a second antenna structure including a second ground and a second feeding portion disposed on the second side wall structure of the second housing, wherein the first side wall structure of the first housing and the second side wall structure of the second housing may include a first metal material, and the first bracket of the first housing includes the first metal material, and the second bracket of the second housing may include a second metal material.

Advantageous Effects

In an electronic device according to various embodiments of the disclosure, based on the hinge structure, the side wall structure and the bracket included in the housing structure in which a heat source is mounted are formed of the same metal material having high thermal conductivity, and the side wall structure and the bracket included in the housing structure in which a heat source is not mounted are at least partially formed of a metal material having a low specific gravity. Thus, it is possible to reduce the weight of the electronic device.

In the electronic device according to various embodiments of the disclosure, it is possible to transfer heat

generated by an electronic component to the outside of a hinge module or the electronic device.

In addition, various effects directly or indirectly understood through this document can be provided.

Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

BRIEF DESCRIPTION OF DRAWINGS

The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;

FIG. 2 is a view illustrating an electronic device in an unfolded state according to an embodiment of the disclosure;

FIG. 3 is a view illustrating an electronic device of FIG. 2 in a folded state according to an embodiment of the disclosure;

FIG. 4 is an exploded perspective view of an electronic device according to an embodiment of the disclosure;

FIG. 5 is a projection view illustrating an internal structure of an electronic device in an unfolded state according to an embodiment of the disclosure;

FIG. 6 is a view illustrating an antenna structure of an electronic device of FIG. 5 in a folded state according to an embodiment of the disclosure;

FIG. 7 is a cross-sectional view of an electronic device of FIG. 5, taken in a direction A-A' in FIG. 5 according to an embodiment of the disclosure;

FIG. 8 is a view illustrating an embodiment in which a second housing structure of an electronic device of FIG. 5 is formed of different metal materials according to an embodiment of the disclosure;

FIGS. 9A and 9B are views each illustrating a ground portion of a first antenna in a first housing structure of an electronic device of FIG. 8 according to an embodiment of the disclosure;

FIGS. 10A and 10B are views each illustrating a ground portion of a second antenna in a second housing structure of an electronic device of FIG. 8 according to an embodiment of the disclosure;

FIG. 11 is an enlarged view of a region A of a first side wall structure of an electronic device of FIG. 8 according to an embodiment of the disclosure;

FIG. 12 is an enlarged view of a region B of a second side wall structure of an electronic device of FIG. 8 according to an embodiment of the disclosure;

FIG. 13 is a cross-sectional view of an electronic device of FIG. 11, taken in a B-B' direction according to an embodiment of the disclosure;

FIG. 14 is a cross-sectional view of an electronic device of FIG. 12, taken in a C-C' direction according to an embodiment of the disclosure;

FIG. 15 is a cross-sectional view of an electronic device of FIG. 12, taken in a D-D' direction according to an embodiment of the disclosure;

FIG. 16 is a view illustrating an embodiment in which a second housing structure of an electronic device of FIG. 5 is formed of a same metal material according to an embodiment of the disclosure; and

FIG. 17 is a cross-sectional view of an electronic device of FIG. 16, taken in an E-E' direction according to an embodiment of the disclosure.

Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

MODE FOR CARRYING OUT THE INVENTION

The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

Referring to FIG. 1, an electronic device 101 in a network environment 100 may communicate with an external electronic device 102 via a first network 198 (e.g., a short-range wireless communication network), or an external electronic device 104 or a server 108 via a second network 199 (e.g., a long-range wireless communication network). According to an embodiment of the disclosure, the electronic device 101 may communicate with the external electronic device 104 via the server 108. According to an embodiment, the electronic device 101 may include a processor 120, memory 130, an input module 150, a sound output module 155, a display module 160, an audio module 170, a sensor module 176, an interface 177, a connecting terminal 178, a haptic module 179, a camera module 180, a power management module 188, a battery 189, a communication module 190, a subscriber identification module (SIM) 196, or an antenna module 197. In some embodiments, at least one of the components (e.g., the connecting terminal 178) may be omitted from the electronic device 101, or one or more other components may be added in the electronic device 101. In some embodiments, some of the components (e.g., the sensor module 176, the camera module 180, or the antenna module 197) may be implemented as a single component (e.g., the display module 160).

The processor 120 may execute, for example, software (e.g., a program 140) to control at least one other component (e.g., a hardware or software component) of the electronic device 101 coupled with the processor 120, and may perform various data processing or computation. According to

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one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in a volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in a non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

The auxiliary processor **123** may control, for example, at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., a sleep) state, or together with the main processor **121** while the main processor **121** is in an active (e.g., executing an application) state. According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of

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the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or an external electronic device (e.g., the external electronic device **102** (e.g., a speaker or a headphone)) directly or wirelessly coupled with the electronic device **101**.

The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the external electronic device **102**) directly or wirelessly. According to an embodiment of the disclosure, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the external electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the external electronic device **102**, the external electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the external electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for

implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

The antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more external devices of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra

low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., a smart home, a smart city, a smart car, or healthcare) based on 5G communication technology or IoT-related technology.

The electronic device according to various embodiments of the disclosure may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B, or C”, “at least one of A, B, and C”, and “at least one of A, B, or C”, may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd”, or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with”, “coupled to”, “connected with”, or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic”, “logic block”, “part”, or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., an internal memory **136** or an external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code

executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

According to various embodiments of the disclosure, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments of the disclosure, one or more of the above-described components or operations may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments of the disclosure, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

FIG. 2 is a view illustrating an electronic device in an unfolded state according to an embodiment of the disclosure. FIG. 3 is a view illustrating an electronic device of FIG. 2 in a folded state according to an embodiment of the disclosure.

In the following detailed description, a pair of housing structures (or referred to as “housings”) may be exemplified with respect to a configuration in which the housing structures are rotatably coupled to each other by a hinge structure (or, referred to as a “hinge module”). However, it should be noted that these embodiments do not limit an electronic device according to various embodiments of the disclosure. For example, an electronic device according to various embodiments of the disclosure may include three or more housing structures, and “a pair of housing structures” in the embodiments disclosed below means “two housing structures rotatably coupled to each other among three or more housing structures”.

Referring to FIGS. 2 and 3, an electronic device **101** (e.g., the electronic device **101** in FIG. 1) may include a pair of housing structures **210** and **220** rotatably coupled to each other via a hinge structure (e.g., a hinge structure **264** in FIG. 4) so as to be folded with respect to each other, a hinge cover

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265 that covers the foldable portions of the pair of housing structures 210 and 220, and a display 230 (e.g., a flexible display or a foldable display) disposed in a space defined by the pair of housing structures 210 and 220. According to an embodiment, the electronic device 101 may include a foldable housing in which the pair of housing structures 210 and 220 are coupled to be rotatable from a position at which the pair of housing structures 110 and 120 are folded to face each other to a position at which the pair of housing structures 210 and 220 are unfolded to be parallel to each other. The surface on which the display 230 according to various embodiments of the disclosure is may be defined as the front surface of the electronic device 101, and the surface opposite to the front surface may be defined as the rear surface of the electronic device 101. In addition, the surface surrounding the space between the front surface and the rear surface may be defined as the side surface of the electronic device 101.

According to an embodiment, the pair of housing structures 210 and 220 may include a first housing structure 210 including a sensor region 231d, a second housing structure 220, a first rear cover 240, and a second rear cover 250. The pair of housing structures 210 and 220 of the electronic device 101 are not limited to the shape and assembly illustrated in FIGS. 2 and 3, but may be implemented by other combinations and/or assemblies of other shapes or components. For example, according to another embodiment, the first housing structure 210 and the first rear cover 240 may be integrally configured, and the second housing structure 220 and the second rear cover 250 may be integrally configured. In another embodiment, the first housing structure 210 may include a first rear cover 240, and the second housing structure 220 may include a second rear cover 250.

According to an embodiment, the first housing structure 210 and the second housing structure 220 may be disposed on opposite sides about a folding axis A, and may have generally symmetrical shapes with respect to the folding axis A. According to some embodiments, the first housing structure 210 and the second housing structure 220 are rotatable with respect the hinge structure 264 or the hinge cover 265 about different folding axes. For example, the first housing structure 210 and the second housing structure 220 may be coupled to the hinge structure 264 or the hinge cover 265 to be individually rotatable. Since the first housing structure 210 and the second housing structure 220 rotate about the folding axis A or about different folding axes, respectively, the first housing structure and the second housing structure are rotatable between the positions at which the first housing structure and the second housing structure are folded to each other and the positions at which the first housing structure and the second housing structure are inclined with respect to each other or the positions at which the first housing structure and the second housing structure are unfolded to be parallel to each other.

In the disclosure, the wording “positioned side by side” or “extending side by side” may mean the state in which two structures (e.g., the housing structures 210 and 220) are at least partially positioned next to each other or the state in which at least portions positioned next to each other are arranged parallel to each other. According to some embodiments, the wording “arranged side by side” may mean that the two structures are arranged to look in parallel or in the same direction while being located next to each other. Expressions such as “side by side”, “parallel”, and the like may be used in the following detailed description, which

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will be readily understood according to the shapes or arrangements of the structures with reference to the accompanying drawings.

According to an embodiment, the angle or distance between the first housing structure 210 and the second housing structure 220 may vary depending on whether the electronic device 101 is in the unfolded state (an extended state, a flat state or an open state), in the folded state (or a closed), or in the intermediate state. In describing various embodiments of the disclosure, when the state of the electronic device 101 is the “unfolded state”, it may mean that the electronic device 101 is in a “fully unfolded state” in which the first housing structure 210 and the second housing structure 220 of the electronic device 101 form 180 degrees therebetween. When the state of the electronic device 101 is the “closed state”, it may mean that the electronic device 101 is in the state in which the first housing structure 210 and the second housing structure 220 of the electronic device 101 form an angle of 0 degrees or less than 10 degrees therebetween. When the state of the electronic device 101 is the “intermediate state”, it means that the angle between the first housing structure 210 and the second housing structure 220 forms an angle between the angle formed in the “unfolded state” and the angle formed in the “closed state”.

According to an embodiment, unlike the second housing structure 220, the first housing structure 210 may further include the sensor region 324 in which various sensors are disposed. However, the first housing structure 210 and the second housing structure 220 may have mutually symmetrical shapes in other regions. According to another embodiment, the sensor region 231d may be additionally disposed or replaced in at least a partial region of the second housing structure 220. According to another embodiment, the sensor region 231d may be omitted from the first housing structure 210.

According to an embodiment, in the unfolded state of the electronic device 101, the first housing structure 210 may include a first surface 211 connected to a hinge structure (e.g., the hinge structure 264 in FIG. 4) and arranged to face the front side of the electronic device 101, a second surface 212 facing away from the first surface 211, and a first side wall structure 213 surrounding at least a portion of the space between the first surface 211 and the second surface 212. According to an embodiment, the first side wall structure 213 may include a first side wall 213a arranged in parallel to the folding axis A, a second side wall 213b extending from one end of the first side wall 213a in a direction perpendicular to the folding axis A, and a third side wall 213c extending from the other end of the first side wall 213a in a direction perpendicular to the folding axis A. In describing various embodiments, expressions such as “parallel” or “perpendicular” are used in connection with the arrangements of the above-described side faces, but in some embodiments, the expressions may include the meaning of “partially parallel” or “partially perpendicular”. In some embodiments, expressions such as “parallel” or “perpendicular” may include an inclined arrangement relationship in an angular range within 10 degrees.

According to an embodiment, in the unfolded state of the electronic device 101, the second housing structure 220 may include a third surface 221 connected to the hinge structure (e.g., the hinge structure 260 in FIG. 4) and arranged to face the front side of the electronic device 101, a fourth surface 222 facing away from the third surface 221, and a second side wall structure 223 surrounding at least a portion of the space between the third surface 221 and the fourth surface 222. According to an embodiment, the second side wall

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structure **223** may include a fourth side wall **223a** arranged in parallel to the folding axis A, a fifth side wall **223b** extending from one end of the fourth side wall **223a** in a direction perpendicular to the folding axis A, and a sixth side wall **223c** extending from the other end of the fourth side wall **223a** in a direction perpendicular to the folding axis A. According to an embodiment, in the folded state, the third surface **221** may be disposed to face the first surface **211**. According to some embodiments, although there are some differences in specific shape, the second side wall structure **223** may be manufactured in substantially the same shape or made of substantially the same material as the first side wall structure **213**.

According to an embodiment, the electronic device **101** may include a recess **201** defined to accommodate the display **230** through structural shape coupling between the first housing structure **210** and the second housing structure **220**. The recess **201** may have substantially the same size as the display **230**.

According to an embodiment, at least a portion of the first housing structure **210** and at least a portion of the second housing structure **220** may be formed of a metal material or a non-metal material having rigidity, the level of which is selected in order to support the display **230**. According to another embodiment, the first housing structure **210** and the second housing structure **220** may at least partially include a conductive material. When the first housing structure **210** and the second housing structure **220** include a conductive material, the electronic device **101** may transmit/receive radio waves by using the portions formed of the conductive material in the first housing structure **210** and the second housing structure **220**. For example, a processor or a communication module of the electronic device **101** may perform wireless communication using a portion of each of the first housing structure **210** and the second housing structure **220**.

According to an embodiment, the sensor region **231d** may be disposed in the display **230** to have a predetermined region without a separate notch adjacent to one corner of the first housing structure **210**. However, the arrangement, shape, and size of the sensor region **231d** is not limited to the illustrated example. For example, in another embodiment, the sensor region **231d** may be provided at another corner of the first housing structure **210** or in any region between the upper and lower end corners. According to another embodiment, the sensor region **231d** may be disposed in at least a partial region of the second housing structure **220**. According to another embodiment, the sensor region **231d** may be disposed to extend over the first housing structure **210** and the second housing structure **220**. According to an embodiment, the electronic device **101** may include components exposed to the front surface of the electronic device **101** through the display **230** or through one or more openings provided in the display **230**, and may perform various functions by using these components. For example, a front camera device disposed in the sensor region **231d** may be exposed on the front surface of the electronic device **101** through the one or more openings provided in the display **230**. However, the disclosure is not necessarily limited to this embodiment, and at least one of the other components disposed in the sensor region **231d**, for example, a proximity sensor, an illuminance sensor, an iris recognition sensor, an ultrasonic sensor, or an indicator, may be exposed on the front surface of the electronic device **101**. The sensor region **231d** may be omitted according to an embodiment, and according to this, the components disposed in the sensor region **231d** may be distributed and disposed in at least a

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portion of the first housing structure **210** and/or at least a portion of the second housing structure **220**.

According to an embodiment, the first rear cover **240** may be disposed on the second face **212** of the first housing structure **210**, and may have a substantially rectangular periphery. According to an embodiment, the periphery of the first rear cover **240** may be at least partially wrapped by the first housing structure **210**. Similarly, the second rear cover **250** may be disposed on the fourth surface **222** of the second housing structure **220**, and at least a portion of the periphery of the second rear cover **250** may be at least partially enclosed by the second housing structure **220**.

According to the illustrated embodiment, the first rear cover **240** and the second rear cover **250** may have substantially symmetrical shapes with respect to the folding axis A. According to another embodiment, the first rear cover **240** and the second rear cover **250** may have various different shapes. According to a still another embodiment, the first rear cover **240** may be formed integrally with the first housing structure **210**, and the second rear cover **250** may be formed integrally with the second housing structure **220**.

According to an embodiment, the first rear cover **240**, the second rear cover **250**, the first housing structure **210**, and the second housing structure **220** may provide, through a mutually coupled structure, a space in which various components (e.g., a printed circuit board, an antenna module, a sensor module, or a battery) of the electronic device **101** may be disposed. According to an embodiment, one or more components may be disposed or visually exposed on the rear surface of the electronic device **101**. For example, one or more components or sensors may be visually exposed through a first rear region **241** of the first rear cover **240**. According to various embodiments, the sensors may include a proximity sensor, a rear camera device, and/or a flash. In an embodiment, at least a portion of the sub-display **252** may be visually exposed through a second rear region **251** of the second rear cover **250**. For example, the sub-display **252** may include the entire second rear region **251** of the second rear cover **250**.

The display **230** may be disposed in a space defined by the pair of housing structures **210** and **220**. For example, the display **230** may be seated in the recess (e.g., the recess **201** in FIG. 2) defined by the pair of housing structures **210** and **220**, and may be disposed to occupy substantially the majority of the front surface of the electronic device **101**. For example, the front surface of the electronic device **101** may include the display **230**, and a partial region (e.g., a peripheral region) of the first housing structure **210** and a partial region (e.g., a peripheral region) of the second housing structure **220**, which are adjacent to the display **230**. According to an embodiment, the rear surface of the electronic device **101** may include a first rear cover **240**, a partial region (e.g., a peripheral region) of the first housing structure **210** adjacent to the first rear cover **240**, a second rear cover **250**, and a partial region (e.g., a peripheral region) of the second housing structure **220** adjacent to the second rear cover **250**.

According to an embodiment, the display **230** may refer to a display, at least a portion of which is deformable into a planar surface or a curved surface. According to an embodiment, the display **230** may include a folding region **231c**, a first region **231a** disposed on one side of the folding region **231c** (e.g., the right region of the folding region **231c**), and a second region **231b** disposed on the other side of the folding region **231c** (e.g., the left region of the folding region **231c**). For example, the first region **231a** may be disposed in the first surface **211** of the first housing structure

210, and the second region **231b** may be disposed in the third surface **221** of the second housing structure **220**. For example, the display **230** may extend from the first surface **211** to the third surface **221** across the hinge structure **264** in FIG. 4, and at least a region corresponding to the hinge structure **264** (e.g., the folding region **231c**) may be a flexible region that is deformable from a flat plate shape into a curved shape.

According to an embodiment, the region division of the display **230** is exemplary, and the display **230** may be divided into multiple regions (e.g., four or more regions or two regions) depending on the structure or functions thereof. For example, in the embodiment illustrated in FIG. 2, the folding region **231c** extends in the direction of the vertical axis (e.g., the Y axis in FIG. 4) parallel to the folding axis A, and the region of the display **230** may be divided by the folding region **231c** or the folding axis (the axis A). However, according another embodiment, the region of the display **230** may be divided with reference to another folding region (e.g., a folding region parallel to the horizontal axis (e.g., the X axis in FIG. 4)) or another folding axis (e.g., a folding axis parallel to the X axis in FIG. 4). The aforementioned region division of the display is merely physical division based on the pair of housing structures **210** and **220** and the hinge structure (e.g., the hinge structure **264** in FIG. 4), and the display **230** may display one full screen substantially through the pair of housing structures **210** and **220** and the hinge structure (e.g., the hinge structure **264** in FIG. 4). According to an embodiment, the first region **231a** and the second region **231b** may have generally symmetrical shapes about the folding region **231c**.

Further referring to FIGS. 2 and 3, the hinge cover **265** may be disposed between the first housing structure **210** and the second housing structure **220** to cover internal components (e.g., the hinge structure **264** in FIG. 4). According to an embodiment, the hinge cover **265** may be covered by a portion of each of the first and second housing structures **210** and **220** or may be exposed to the outside depending on the operating state of the electronic device **101** (the unfolded state or the folded state).

Hereinafter, the operations of the first housing structure **210** and the second housing structure **220** and respective regions of the display **230** depending on the operating state of the electronic device **101** (e.g., the unfolded state (extended state) and the folded state) will be described.

According to an embodiment, when the electronic device **101** is in the unfolded state (extended state) (e.g., the state in FIG. 2), the first housing structure **210** and the second housing structure **220** may form an angle of 180 degrees therebetween, and the first region **231a** and the second region **231b** of the display may be disposed to face the same direction, for example, to display screens in directions parallel to each other. In addition, the folding region **231c** may form the same plane as the first region **231a** and the second region **232b**.

According to an embodiment, when the electronic device **101** is in the folded state (e.g., the state of FIG. 3), the first housing structure **210** and the second housing structure **220** may be disposed to face each other. For example, when the electronic device **101** is in the folded state (e.g., the state of FIG. 3), the first region **231a** and the second region **231b** of the display **230** may form a narrow angle (e.g., 0 to 10 degrees) therebetween and may face each other. When the electronic device **101** is in the folded state (e.g., the state of FIG. 3), at least a portion of the folding region **231c** may form a curved surface having a predetermined curvature.

According to an embodiment, when the electronic device **101** is in the intermediate state, the first housing structure **210** and the second housing structure **220** may be disposed to form therebetween a predetermined angle of, for example, 90 degrees or 120 degrees. For example, in the intermediate state, the first region **231a** and the second region **231b** of the display **230** may form an angle larger than that in the folded state and smaller than that in the unfolded state. At least a portion of the folding region **231c** may be formed as a curved surface having a predetermined curvature, and the curvature in this case may be smaller than that in the folded state.

FIG. 4 is an exploded perspective view of an electronic device according to an embodiment of the disclosure.

FIG. 4 illustrates a direction component X, a direction component Y, and a direction component Z. According to an embodiment of the disclosure, the direction component X, the direction component Y, and the direction component Z may represent, respectively, the X axis, the Y axis, and the Z axis of a spatial coordinate system. In various embodiments of the disclosure, the X axis may indicate the width direction of the electronic device, the Y axis may indicate the length direction of the electronic device of the electronic device, and the Z axis may indicate the height direction (or thickness direction) of the electronic device.

Referring to FIG. 4, according to an embodiment, the electronic device **101** (e.g., the electronic device **101** in FIGS. 1 to 3) may include a display **230**, a bracket assembly **260**, at least one printed circuit board **270** (e.g., a PCB, a flexible PCB (FPCB), or a rigid flexible PCB (RFPCB)), a first housing structure **210**, a second housing structure **220**, a first rear cover **240**, and a second rear cover **250**. The display **230** according to various embodiments of the disclosure may be referred to as a display module (e.g., the display module **160** in FIG. 1) or a display assembly.

The display **230** may include a display panel **231** (e.g., a flexible display panel), and at least one plate **232** or layer on which the display panel **231** is seated. According to an embodiment, the plate **232** may be disposed between the display panel **231** and the bracket assembly **260**. According to an embodiment, the plate **232** may configure at least a portion of the first housing structure **210** and the second housing structure **220**. The display panel **231** may be disposed on at least portion of one surface (e.g., the surface in the Z direction in FIG. 4) of the plate **232**. The plate **232** may be formed in a shape corresponding to that of the display panel **231**.

The bracket assembly **260** may include a first bracket **261**, a second bracket **262**, a hinge structure **264** disposed between the first bracket **261** and the second bracket **262**, a hinge cover **265** configured to cover the hinge structure **264** when the hinge structure **264** is viewed from the outside, and a wiring member **263** (e.g., a flexible printed circuit board (FPCB)) extending across the first and second brackets **261** and **262**.

According to an embodiment, the bracket assembly **260** may be disposed between the plate **232** and the at least one printed circuit board **270**. For example, the first bracket **261** may be disposed between the first region **231a** of the display **230** and a first printed circuit board **271**. The second bracket **262** may be disposed between the second region **231a** of the display **230** and a second printed circuit board **272**.

According to an embodiment, at least a portion of the wiring member **263** and the hinge structure **264** may be disposed inside the bracket assembly **260**. The wiring member **263** may be disposed in a direction across the first bracket **261** and the second bracket **262** (e.g., the x-axis

direction). The wiring member **263** may be disposed in a direction perpendicular to the folding axis (e.g., the Y axis or the folding axis A in FIG. 2) of the folding region **231c** (e.g., the X-axis direction).

According to various embodiments, the hinge structure **264** may include a hinge module **264a**, a first hinge plate **264b**, and/or a second hinge plate **264c**. According to some embodiments, the hinge module **264a** may be interpreted as including the first hinge plate **264b** and the second hinge plate **264c**. According to an embodiment, the first hinge plate **264b** may be mounted inside the first housing structure **210**, and the second hinge plate **264c** may be mounted inside the second housing structure **220**. According to some embodiments, the first hinge plate **264b** may be coupled to the first bracket **261**, and the second hinge plate **264c** may be coupled to the second bracket **262**. According to another embodiment, the first hinge plate **264b** (or the second hinge plate **264c**) may be coupled to another structure (e.g., a first rotation support surface **214** or a second rotation support surface **224**) inside the first housing structure **210** (or the second housing structure **220**). For example, the structure to which the first hinge plate **264b** (or the second hinge plate **264c**) is coupled inside the first housing structure **210** (or the second housing structure **220**) may vary according to embodiments. According to another embodiment, the hinge module **264a** may be coupled to the first hinge plate **264b** and the second hinge plate **264c** to rotatably connect the second hinge plate **264c** to the first hinge plate **264b**. For example, a folding axis (e.g., the folding axis A in FIG. 2) is provided by the hinge module **264a**, and the first housing structure **210** and the second housing structure **220** (or the first bracket **261** and the second bracket **262**) may rotate with respect to each other substantially about the folding axis A.

As described above, the at least one printed circuit board **270** may include a first printed circuit board **271** disposed on the first bracket **261** side and a second printed circuit board **272** disposed on the second bracket **262** side. The first printed circuit board **271** and the second printed circuit board **272** may be disposed inside a space defined by the bracket assembly **260**, the first housing structure **210**, the second housing structure **220**, the first rear cover **240**, and the second rear cover **250**. Components for implementing various functions of the electronic device **101** may be mounted on the first printed circuit board **271** and the second printed circuit board **272**.

According to an embodiment, the first housing structure **210** and the second housing structure **220** may be assembled so as to be coupled to the opposite sides of the bracket assembly **260** in the state in which the display **230** is coupled to the bracket assembly **260**. The first housing structure **210** and the second housing structure **220** may be slidably coupled to the opposite sides of the bracket assembly **260**, for example, to the first bracket **261** and the second bracket **262**, respectively. The first bracket **261** and the second bracket **262** are substantially accommodated in the first housing structure **210** and the second housing structure **220**. According to an embodiment, the first bracket **261** and the second bracket **262** may be interpreted as portions of the first housing structure **210** and the second housing structure **220**, respectively.

According to an embodiment, the first housing structure **210** may include a first rotation support surface **214**, and the second housing structure **220** may include a second rotation support surface **224**, which corresponds to the first rotation support structure **214**. The first rotation support surface **214** and the second rotation support surface **224** may include curved surfaces corresponding, respectively, to curved sur-

faces included in the hinge cover **265**. According to an embodiment, the first housing structure **210** may include at least one of a plurality of components **280** disposed in the sensor region **231d** of FIGS. 2 and 3 (e.g., a front camera device, a proximity sensor, an illuminance sensor, an iris recognition sensor, an ultrasonic sensor, or an indicator).

In an embodiment, when the electronic device **101** is in the unfolded state (e.g., the state of FIG. 2), the first rotation support surface **214** and the second rotation support surface **224** cover the hinge cover **265**, so that the hinge cover **265** may not be exposed or may be minimally exposed to the rear surface of the electronic device **101**. In an embodiment, when the electronic device **101** is in the unfolded state (e.g., the state of FIG. 3), the first rotation support surface **214** and the second rotation support surface **224** rotate, respectively, along the curved surfaces included in the hinge cover **265**, so that the hinge cover **265** can be maximally exposed to the rear surface of the electronic device **101**.

In the foregoing detailed description, in the first housing structure **210**, the second housing structure **220**, the first side wall structure **213**, the second side wall structure **223**, and the like, ordinal numbers are used merely to distinguish components. It is noted that the disclosure is not limited by the description of the ordinal numbers. For example, although the sensor region **231d** is illustrated as being provided in the first housing structure **210**, the sensor region **231d** may be provided in the second housing structure **220** or may be provided in both the first and second housing structures **210** and **220**. In another embodiment, a configuration in which the first rear region **241** and the sub-display **252** are disposed on the first rear cover **240** and the second rear cover **250**, respectively, is exemplified, but both the first rear region **241** for disposing a sensor or the like and the sub-display **252** for outputting a screen may be disposed on one of the first rear cover **240** and the second rear cover **250**.

FIG. 5 is a projection view illustrating an internal structure of an electronic device in a unfolded state according to an embodiment of the disclosure. FIG. 6 is a view illustrating an antenna structure of an electronic device of FIG. 5 according to an embodiment of the disclosure. FIG. 7 is a cross-sectional view of an electronic device of FIG. 5, taken in the direction A-A' in FIG. 5 according to an embodiment of the disclosure.

Referring to FIG. 5, the electronic device **101** (e.g., the electronic device **101** in FIGS. 1 to 4) may be provided with a foldable housing including a folding region H so that the electronic device **101** is substantially folded about a folding axis, wherein the foldable housing includes a first side wall structure **510** (e.g., the first side wall structure **213** in FIG. 4) included in a first housing structure **410** (e.g., the first housing structure **210** in FIGS. 2 to 4) and a second side wall structure **520** (e.g., the second side wall structure **223** in FIGS. 2 to 4) included in a second housing structure **420** (e.g., the second housing structure **220** in FIGS. 2 to 4).

The foldable housing includes at least one printed circuit board (e.g., the at least one printed circuit board **270** in FIG. 4). According to an embodiment, in a region or space of the first housing structure **410** that is surrounded by the first side wall structure **510**, a first printed circuit board **710** (e.g., the first printed circuit board **271** in FIG. 4) and a second printed circuit board **720** (e.g., the first printed circuit board **271** in FIG. 4) may be accommodated. According to an embodiment, in a region or space of the second housing structure **420** that is surrounded by the second side wall structure **520**, a third printed circuit board **730** (e.g., the second printed

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circuit board **272** in FIG. **4**) and a fourth printed circuit board **740** (e.g., the second printed circuit board **272** in FIG. **4**) may be accommodated.

According to an embodiment, the first printed circuit board **710** may be a main PCB that includes a processor **713** (e.g., the processor **120** in FIG. **1**), a power management module (e.g., a PMIC), a charger IC, a wireless communication circuit, or a wireless transceiver (a radio frequency transceiver). According to an embodiment, the first printed circuit board **710** may further include a first communication device **714** and/or a second communication device **715**. According to another embodiment, the first communication device **714** and/or the second communication device **715** may be provided in a separate module form. For example, the first communication device **714** and/or the second communication device **715** may be a mmWave communication device that performs wireless communication in a frequency band of 20 GHz or higher and 100 GHz or lower. According to an embodiment, the processor **713** may include at least a communication processor (CP) or a component in which an application processor (AP) and a communication processor are integrated, and may control or drive a wireless transceiver, a power management module, a wireless communication circuit, or the like. According to an embodiment, the wireless transceiver may communicate with the processor **713** with a transmission/reception signal and/or a control signal. In another embodiment, the wireless transceiver may be included in the third printed circuit board **730** included in the second housing structure **420**. According to an embodiment, the second printed circuit board **720** may be a universal serial bus (USB) port **721**, a speaker device, or a sub-PCB including an antenna. According to an embodiment, the speaker device or the antenna may be provided in the form of a separate module and connected to the second printed circuit board **720**.

According to an embodiment, various components (e.g., a first camera device **711** and a second camera device **712**) may be additionally provided on the first printed circuit board **710**, and various components (e.g., a microphone **722**) may be additionally provided to the second printed circuit board **720** as well. In some embodiments, the camera devices **711** and **712** may be disposed adjacent to the first printed circuit board **710** in the state being separated from the first printed circuit board **710**. It should be noted that the number and arrangement of various components in various embodiments of the disclosure are not limited to the embodiment illustrated in FIG. **5** and may be set variously according to an embodiment.

According to an embodiment, the third printed circuit board **730** may be a main PCB including a third communication device **733**. According to another embodiment, the third communication device **733** may be provided in the form of a separate module. For example, the third communication device **733** may be a Wi-Fi communication device that performs wireless communication in a frequency band of 900 MHz, 2.4 GHz, 3.65 Hz, 4.9 to 5.0 GHz, or 5 GHz or higher. According to an embodiment, the fourth printed circuit board **740** may be a sub-PCB including a speaker device or an antenna. According to an embodiment, the speaker device or the antenna may be provided in the form of a separate module and connected to the fourth printed circuit board **740**.

According to an embodiment, various components (e.g., a third camera device **732** and a receiver **731**) may be additionally provided on the third printed circuit board **730**, and various components may be additionally provided on the fourth printed circuit board **740** as well. In some embodi-

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ments, the third camera device **732** may be disposed adjacent to the third printed circuit board **730** in the state being separated from the third printed circuit board **730**. It should be noted that the number and arrangement of various components in various embodiments of the disclosure are not limited to the embodiment illustrated in FIG. **5** and may be set variously according to an embodiment.

According to an embodiment, the printed circuit boards **710**, **720**, **730**, and **740** may include a plurality of connectors **C** for electrical connection (e.g., control signal, power, or communication signal transmission) between the various components or for electrical connection among the printed circuit boards **710**, **720**, **730**, and **740**. In the plurality of connectors **C** included in the printed circuit boards **710**, **720**, **730**, and **740**, various types of connection structures (or connector structures), such as a flexible printed circuit (FPC) or flexible flat cable (FFC) type, a board to board (B to B) type, a zip type, a bonding type provided through a hot bar process, a low insertion force (LIF), a zero insertion force (ZIF) and the like may be adopted, and through these, relevant components may be electrically coupled to each other or respective printed circuit boards may be electrically coupled to each other.

According to an embodiment, in the electronic device **101**, an electrical connection structure **F** (or a connector structure) of a connector to connector type (or C2C type) that interconnects two different connectors **C** to each other may be adopted in addition to the various types of connection structures.

According to an embodiment, the first printed circuit board **710** and the second printed circuit board **720** may be configured to be supported from a first bracket **610** (e.g., the first bracket **261** in FIG. **4**) coupled to the first side wall structure **510**. According to an embodiment, the third printed circuit board **730** and the fourth printed circuit board **740** may be configured to be supported from a second bracket **620** (e.g., the second bracket **262** in FIG. **4**) coupled to the second side wall structure **520**. According to an embodiment, the first side wall structure **510** and the first bracket **610** may be configured separably or integrally. According to an embodiment, the second side wall structure **520** and the second bracket **620** may be configured separably or integrally. According to an embodiment, the first bracket **610** may support the first printed circuit board **710** and the second printed circuit board **720**, or may support other components (e.g., a battery **751**) that are not mounted on the first printed circuit board **710** and the second printed circuit board **720**. According to an embodiment, the second bracket **620** may support the third printed circuit board **730** and the fourth printed circuit board **740**, or may support other components (e.g., a battery **752** and a SIM reader **761**) that are not mounted on the third printed circuit board **730** and the fourth printed circuit board **740**.

According to an embodiment, the first housing structure **410** may further include a wireless charging module (not illustrated) equipped with a wireless charging part, a near field communication (NFC) antenna, and/or a magnetic secure transmission (MST) antenna. For example, the wireless charging module may be mounted on the first battery **751** supported by the first bracket **610** in the first housing structure **410**.

Referring to FIG. **6**, the first housing structure **410** of the electronic device **101** includes antenna regions (e.g., a (1-1)th antenna region **811**, a (1-2)th antenna region **812**, a (1-3)th antenna region **813**, a (1-4)th antenna region **814**, a (1-5)th antenna region **815**, a (1-6)th antenna region **816**, a (1-7)th antenna region **817**, and a (1-8)th antenna region **818**

in each of which an antenna is disposed. For example, a (1-1)th antenna operating in an LB frequency band as a first main antenna may be disposed in the (1-1)th antenna region **811**, a (1-2)th antenna operating in an MB, HB, n77, or n78 band as a second main antenna may be disposed in the (1-2)th antenna region **812**, a (1-3)th antenna operating in an n41 or n79 band as a third main antenna may be disposed in the (1-3)th antenna region **813**, and a (1-4)th antenna operating in an mb band as a fourth main antenna may be disposed in the (1-4)th antenna region **814**. A (1-5)th antenna operating in an lb, mb, hb, or n41 frequency band as a first sub-antenna may be disposed in the (1-5)th antenna region **815**, a (1-6)th antenna operating in a 77, N78, n79, LAA1, L1, mb, or hb band as a second sub-antenna may be disposed in the (1-6)th antenna region **816**, a (1-7)th antenna operating in an N79, n77, n78, L5, or an ultra-wide (UWB9) band as a third sub-antenna may be disposed in the (1-7)th antenna region **817**, and a (1-8)th antenna operating in an n77, n78, n79, or LAA2 band as a fourth sub-band may be disposed in the (1-8)th antenna region **818**. The number of antenna regions included in the first housing structure **410** is limited to eight in FIG. 6, the number and locations of antenna regions are not limited thereto and may be changed as necessary.

According to an embodiment, the second housing structure **420** of the electronic device **101** may include antenna regions (e.g., a (2-1)th antenna region **821** and a (2-2)th antenna region **822**) in each of which an antenna is disposed. For example, a (2-1)th antenna operating as a first Wi-Fi antenna may be disposed in the (2-1)th antenna region **821**, and a (2-2)th antenna operating as a second Wi-Fi antenna may be disposed in the (2-2)th antenna region **822**. The number of antenna regions included in the second housing structure **420** is limited to two in FIG. 6, the number and locations of antenna regions are not limited thereto and may be changed as necessary.

According to an embodiment, the first side wall structure **510** may include at least one split portion **531**, **532**, **533**, **534**, **535**, or **536**. According to an embodiment, the at least one split portion **531**, **532**, **533**, **534**, **535**, or **536** may be formed of an insulator. According to an embodiment, the at least one split portion **531**, **532**, **533**, **534**, **535**, or **536** may be molded into the first side wall structure **510** through a double-injection molding or insert molding method of a synthetic resin. However, without being limited thereto, the at least one split portion **531**, **532**, **533**, **534**, **535**, or **536** may be made of various materials having insulating properties. According to an embodiment, the first side wall structure **510** may include unit side wall portions **511**, **512**, **513**, **514**, **515**, **516**, and **517** split by respective split portions **531**, **532**, **533**, **534**, **535**, and **536**. For example, a (1-2)th side wall portion **512** may maintain the state of being separated from a (1-1)th side wall portion **511** and a (1-3)th side wall portion **513** by a (1-1)th split portion **531** and a (1-2)th split portion **532**, a (1-4)th side wall portion **514** may maintain the state of being separated from a (1-3)th side wall portion **513** and a (1-5)th side wall portion **515** by a (1-3)th split portion **533** and a (1-4)th split portion **534**, and a (1-6)th side wall portion **516** may maintain the state of being separated from a (1-5)th side wall portion **515** and a (1-7)th side wall portion **517** by a (1-5)th split portion **535** and a (1-6)th split portion **536**. According to an embodiment, at least one of the unit side wall portions **511**, **512**, **513**, **514**, **515**, **516**, and **517** may operate as an antenna operating in a specific frequency band. For example, the (1-1)th side wall portion **511** may operate as the (1-6)th antenna disposed in the (1-6)th antenna region **816**, the (1-2)th side wall portion **512** may operate as the (1-7)th antenna disposed in the (1-7)th antenna region **817**,

the (1-3)th side wall portion **513** may operate as the (1-5)th antenna disposed in the (1-5)th antenna region **815**, the (1-5)th side wall portion **515** may operate as the (1-1)th antenna disposed in the (1-1)th antenna region **811**, the (1-6)th side wall portion **516** may operate as the (1-2)th antenna disposed in the (1-2)th antenna region **812**, and the (1-7)th side wall portion **517** may operate as the (1-4)th antenna disposed in the (1-4)th antenna region **814**. The number of split portions included in the first side wall structure **510** is limited to six in FIG. 6, the number and locations of split portions are not limited thereto and may be changed as necessary.

According to an embodiment, the second side wall structure **520** may include at least one split portion **541**, **542**, **543**, **544**, **545**, or **546**. According to an embodiment, the at least one split portion **541**, **542**, **543**, **544**, **545**, or **546** may be formed of an insulator. According to an embodiment, the at least one split portion **541**, **542**, **543**, **544**, **545**, or **546** may be molded into the second side wall structure **520** through a double-injection molding or insert molding method of a synthetic resin. However, without being limited thereto, the at least one split portion **541**, **542**, **543**, **544**, **545**, or **546** may be made of various materials having insulating properties. According to an embodiment, the second side wall structure **520** may include unit side wall portions **521**, **522**, **523**, **524**, **525**, **526**, and **527** split by respective split portions **541**, **542**, **543**, **544**, **545**, and **546**. For example, a (2-2)th side wall portion **522** may maintain the state of being separated from a (2-1)th side wall portion **521** and a (2-3)th side wall portion **523** by a (2-1)th split portion **541** and a (2-2)th split portion **542**, a (2-4)th side wall portion **524** may maintain the state of being separated from a (2-3)th side wall portion **523** and a (2-5)th side wall portion **525** by a (2-3)th split portion **543** and a (2-4)th split portion **544**, and a (2-6)th side wall portion **526** may maintain the state of being separated from a (2-5)th side wall portion **525** and a (2-7)th side wall portion **527** by a (2-5)th split portion **545** and a (2-6)th split portion **546**. According to an embodiment, at least one of the unit side wall portions **521**, **522**, **523**, **524**, **525**, **526**, and **527** may operate as an antenna operating in a specific frequency band. For example, the (2-1)th side wall portion **521** may operate as the (2-1)th antenna disposed in the (2-1)th antenna region **821**, and the (2-3)th side wall portion **523** may operate as the (2-2)th antenna disposed in the (2-2)th antenna region **822**. The number of split portions included in the second side wall structure **520** is limited to six in FIG. 6, the number and locations of split portions are not limited thereto and may be changed as necessary.

Referring to FIG. 7, a heat dissipation structure **900** may be disposed in the first housing structure **410** of the electronic device **101**. According to an embodiment, the heat dissipation structure **900** may include a first printed circuit board **710**, at least one heating element **910** (e.g., the processor **713** of FIG. 5) disposed on the first printed circuit board **710**, a shield can **920** coupled to at least one surface of the first printed circuit board **710** and including a shape (e.g., a closed tetragonal loop) surrounding at least a portion of the at least one heating element **910**, at least one heat transfer member **930** (e.g., a carbon fiber thermal interface material (TIM)) disposed on the at least one heating element **910** and transferring heat generated by the at least one heating element **910**, a heat diffusion structure **940** disposed over at least one heat transfer member **930** and at least a portion of the shield can **920**, configured to provide a shielding function against electromagnetic waves that may be generated by the at least one heating element **910**, and configured to provide a function of transferring heat capable

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of being generated by the at least one heating element **910** to the outside, at least a portion of the first bracket **610** disposed on the heat diffusion structure **940** and configured to quickly dissipate the heat transferred from the at least one heating element **910** to the outside of the electronic device **101**, and/or a heat diffusion material (e.g., a vapor chamber or a laminated graphite) **950** disposed on the at least a portion of the first bracket **610**. In the illustrated embodiment, the first printed circuit board **710** is exemplified as a single board, but the various embodiments disclosed herein are not limited thereto, and may include a structure in which a plurality of boards are coupled to face each other. When the first printed circuit board **710** has a structure in which a plurality of boards are coupled to face each other, the at least one heating element **910** may be disposed to form a multi-layer with a heating element disposed on another board in the first printed circuit board **710**.

According to an embodiment, the heat diffusion structure **940** may include a heat dissipation member **942** and an elastic member **841** provided along an edge of the heat dissipation member **942**. The heat dissipation member **942** of the heat diffusion structure **940** may be disposed in direct contact with the at least one heat transfer member **930**, and may diffuse the heat transferred from the at least one heating element **910** and transfer the heat to at least a portion of the first bracket **610**. An elastic member **941** of the heat diffusion structure **940** may be connected to the heat dissipation member **942** and disposed on a portion of the shield can **920** to provide an overall elastic force to the heat diffusion structure **940**. A heat transfer path connected from the at least one heating element **910** and/or the shield can **920** to at least a portion of the first bracket **610** may be expanded and the at least one heating element **910** may be protected from an external impact by the elastic force of the elastic member **941**.

An embodiment in which at least a portion of the first bracket **610** and the heat diffusion material **950** are both disposed on the heat diffusion structure **940** have been described above with reference to FIG. 7, but the at least a portion of the first bracket **610** may be omitted and the heat diffusion material **950** may be disposed directly on the heat diffusion structure **940**.

FIG. 8 is a view illustrating an embodiment in which a second housing structure of an electronic device of FIG. 5 is formed of different metal materials according to an embodiment of the disclosure.

Referring to FIG. 8, the first side wall structure **510** included in the first housing structure **410** of the electronic device **101** may include a metal material having high electrical conductivity for the performance of the communication device (e.g., antenna performance). For example, the first side wall structure **510** may include an aluminum (Al) material.

According to an embodiment, the first printed circuit board **710** included in the first housing structure **410** may include a structure in which one or more heating elements (e.g., the processor **713** in FIG. 5, a PMIC, and a charger IC), which are heat sources that generate heat, are arranged in multiple layers (e.g., a plurality of PBAs connected by an interposer). For example, the plurality of heating elements may be disposed on the first printed circuit board **710**. According to an embodiment, the first bracket **610** coupled to the first side wall structure **510** may include a metal material having high thermal conductivity for heat dissipation performance (e.g., heat transfer or emission). For example, the first bracket **610** coupled to the first side wall structure **510** may include an Al material.

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According to an embodiment of the disclosure, the first bracket **610** may include the same metal material as the first side wall structure **510** in order to lower the resistance of a ground (GND) connected to an antenna. For example, the first bracket **610** and the first side wall structure **510** may include an Al material. According to an embodiment, the first side wall structure **510** and the first bracket **610** may be configured integrally.

According to an embodiment, the second housing structure **420** may include a sub-display (e.g., the sub-display **252** in FIGS. 2 to 4) disposed in a region of the rear cover (e.g., the second rear cover **250** in FIGS. 2 to 4). According to an embodiment, the third printed circuit board **730** included in the second housing structure **420** may include a structure in which one or more heating elements (e.g., the third communication device **733** of FIG. 5, a speaker amplifier, a camera filter, and a power module) are arranged in a single layer (e.g., a single-layer PBA structure). For example, by provided in a single-layer PBA structure when the sub-display is disposed in a region of the rear cover, the third printed circuit board **730** may not interfere with a flexible display (e.g., the display **130** in FIG. 4) and/or the sub-display. According to an embodiment, elements that generate relatively less heat compared to the first printed circuit board **710** included in the first housing structure **410** may be disposed on the third printed circuit board **730**.

According to an embodiment, the second side wall structure **520** may include a metal material having high electrical conductivity for performance of a communication device (e.g., antenna performance). For example, the second side wall structure **520** may include an Al material.

According to an embodiment, the second bracket **620** coupled to the second side wall structure **520** may include a metal material having a low specific gravity for weight reduction. For example, in the second housing structure **420**, the third printed circuit board **730** may include elements that generate relatively less heat compared to the first printed circuit board **710** included in the first housing structure **410**. Thus, the degree of design freedom of the second bracket **620** may be higher than that of the first bracket **610** included in the first housing structure **410** in terms of material selection. According to an embodiment, since the second bracket **620** includes a metal material having a low specific gravity, it is possible to reduce the weight increased by the sub-display compared to the first housing structure **410**. For example, the second bracket **620** may include a magnesium (Mg) material having a low specific gravity. According to an embodiment, the second side wall structure **520** and the second bracket **620** may include different metal materials. According to an embodiment, the second side wall structure **520** and the second bracket **620** may be configured separately. For example, the second side wall structure **520** and the second bracket **620** may be assembled or coupled to each other through a bonding method, such as welding.

Referring to FIG. 8, the same metal material is shaded in the same manner for convenience of description. For example, in FIG. 8, it can be seen that the first sidewall structure **510**, the first bracket **610**, and the second side wall structure **520** are shaded in the same way to indicate that they are made of the same metal material (e.g., Al), and the second bracket **620** and the second side wall structure **520** are shaded in different ways to indicate that the second bracket is made of a material (e.g., Mg) different from that of the second side wall structure **520**.

When at least a portion of a side wall structure (e.g., the first side wall structure **510** or the second side wall structure **520**) in the electronic device **101** of FIG. 8 operates as an

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antenna, the ground portion of the antenna may mean a ground portion of components (included in the printed circuit board) directly or indirectly connected to an antenna feeding portion of the antenna among the components included in a printed circuit board (the first printed circuit board 710, the second printed circuit board 720, the third printed circuit board 730, or the fourth printed circuit board 740). Hereinafter, when at least a portion of the first side wall structure 510 operates as a first antenna in the first housing structure 410 in which the first side wall structure 510 and the first bracket 610 include the same metal material, the connection state between the ground portion of the first antenna and the ground portion of the first printed circuit board 710 (or the second printed circuit board 720) will be described with reference to FIGS. 9A and 9B. When at least a portion of the second side wall structure 520 operates as a second antenna in the second housing structure 420 in which the second side wall structure 520 and the second bracket 620 include different metal materials, the connection state between the ground portion of the second antenna and the ground portion of the third printed circuit board 730 (or the fourth printed circuit board 740) will be described with reference to FIGS. 10A and 10B.

FIGS. 9A and 9B are views each illustrating a ground portion of a first antenna 518 in the first housing structure 410 of the electronic device 101 of FIG. 8 according to an embodiment of the disclosure. FIG. 9A illustrates an embodiment in which the ground portion of the first antenna 518 is connected to a ground portion of a component indirectly connected to the feeding portion of the first antenna 518, and FIG. 9B illustrates an embodiment in which the ground portion of the first antenna 518 is connected to a ground portion of a component directly connected to the feeding portion of the first antenna 518.

Referring to FIG. 9A, at least a portion of the first side wall structure 510 of the electronic device 101 (e.g., the (1-1)th side wall portion 511) may operate as the first antenna 518. According to an embodiment, the first antenna 518 may include a first switch portion 518-1, a first feeding portion 518-3, and a first ground portion 518-2. According to an embodiment, the first switch portion 518-1 of the first antenna 518 may be connected to a first component 911 (e.g., a switch) included in the first printed circuit board 710, and the first component 911 may be connected to a ground portion 710-1 of the first printed circuit board 710. According to an embodiment, the first feeding portion 518-3 of the first antenna 518 may be connected to a second component 912 (e.g., a passive element) included in the first printed circuit board 710, and the second component 912 may be connected to the first component 911 and indirectly connected to the ground portion 710-1 of the first printed circuit board 710. According to an embodiment, the ground portion 710-1 of the first printed circuit board 710 may be connected to the first bracket 610 and may be directly connected to the first ground portion 518-2 of the first antenna 518.

Referring to FIG. 9B, at least a portion of the first side wall structure 510 of the electronic device 101 (e.g., the (1-1)th side wall portion 511) may operate as the first antenna 518. According to an embodiment, the first antenna 518 may include a first switch portion 518-1, a first feeding portion 518-3, and a first ground portion 518-2. According to an embodiment, the first switch portion 518-1 of the first antenna 518 may be connected to a third component 913 (e.g., a switch) included in the first printed circuit board 710, and the third component 913 may be connected to the ground portion 710-1 of the first printed circuit board 710. According to an embodiment, the first feeding portion 518-3

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of the first antenna 518 may be connected to the third component 913 included in the first printed circuit board 710. According to an embodiment, the ground portion 710-1 of the first printed circuit board 710 may be connected to the first bracket 610, and the first ground portion 518-2 of the first antenna 518 may be connected to the first bracket 610, so that the first ground portion 518-2 of the first antenna 518 may be indirectly connected to the ground portion 710-1 of the first printed circuit board 710.

An embodiment in which the first ground portion 518-2 of the first antenna 518 is directly or indirectly connected to the ground portion 710-1 of the first printed circuit board 710 has been described with reference to FIGS. 9A and 9B, the embodiment of FIGS. 9A and 9B is also applicable to the second printed circuit board 720 included in the first housing structure 410.

FIGS. 10A and 10B are views each illustrating a ground portion of a second antenna 528 in a second housing structure 420 of an electronic device of FIG. 8 according to an embodiment of the disclosure. FIG. 10A illustrates an embodiment in which a ground portion of a second antenna 528 is connected to a ground portion of a component indirectly connected to a feeding portion of a second antenna 528, and FIG. 10B illustrates an embodiment in which a ground portion of a second antenna 528 is connected to a ground portion of a component directly connected to a feeding portion of a second antenna 528.

Referring to FIG. 10A, at least a portion of the second side wall structure 520 of the electronic device 101 (e.g., the (2-1)th side wall portion 521) may operate as the second antenna 528. According to an embodiment, the second antenna 528 may include a second switch portion 528-1, a second feeding portion 528-3, and a second ground portion 528-2. According to an embodiment, the second switch portion 528-1 of the second antenna 528 may be connected to a fourth component 921 (e.g., a switch) included in the third printed circuit board 730, and the fourth component 921 may be connected to the ground portion 730-1 of the third printed circuit board 730. According to an embodiment, the second feeding portion 528-3 of the second antenna 528 may be connected to a fifth component 922 (e.g., a passive element) included in the third printed circuit board 730, and the fifth component 922 may be connected to the fourth component 921 and indirectly connected to the ground portion 730-1 of the third printed circuit board 730. According to an embodiment, the ground portion 730-1 of the third printed circuit board 730 may be connected to the second bracket 620, and the second ground portion 528-2 of the second antenna 528 may be directly connected to the ground portion 730-1 of the third printed circuit board 730.

Referring to FIG. 10B, at least a portion of the second side wall structure 520 of the electronic device 101 (e.g., the (2-1)th side wall portion 521) may operate as the second antenna 528. According to an embodiment, the second antenna 528 may include a second switch portion 528-1, a second feeding portion 528-3, and a second ground portion 528-2. According to an embodiment, the second switch portion 528-1 of the second antenna 528 may be connected to a sixth component 923 (e.g., a switch) included in the third printed circuit board 730, and the sixth component 923 may be connected to the ground portion 730-1 of the third printed circuit board 730. According to an embodiment, the second feeding portion 528-3 of the second antenna 528 may be connected to the sixth component 923 included in the third printed circuit board 730. According to an embodiment, the ground portion 730-1 of the third printed circuit board 730 may be connected to the second bracket 620, and

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the second ground portion **528-2** of the second antenna **528** may be directly connected to the ground portion **730-1** of the third printed circuit board **730**.

An embodiment in which the second ground portion **528-3** of the second antenna **528** is directly connected to the ground portion **730-1** of the third printed circuit board **730** has been described with reference to FIGS. **10A** and **10B**, the embodiment of FIGS. **10A** and **10B** is also applicable to the fourth printed circuit board **740** included in the second housing structure **420**.

Referring to FIGS. **9A**, **9B**, **10A**, and **10B**, in the case of the first housing structure **410**, since the first side wall structure **510** and the first bracket **610** include the same metal material, the first ground portion **518-2** of the first antenna **518** and the ground portion **710-1** of the first printed circuit board **710** may be directly or indirectly connected to each other, but in the case of the second housing structure **420**, since the second side wall structure **520** and the second bracket **620** include different metal materials, the second ground portion **528-2** of the second antenna **528** and the ground portion **730-1** of the third printed circuit board **730** may only be directly connected to each other.

FIG. **11** is an enlarged view of a region A of a first side wall structure of an electronic device of FIG. **8** according to an embodiment of the disclosure. FIG. **12** is an enlarged view of a region B of a second side wall structure of an electronic device of FIG. **8** according to an embodiment of the disclosure.

Referring to FIG. **11**, the first side wall structure **510** may include a (1-2)th side wall portion **511** separated from a (1-1)th side wall portion and a (1-3)th side wall portion by a (1-1)th split portion **531** and a (1-2)th split portion **532**. According to an embodiment, the (1-2)th side wall portion **512** may operate as an antenna of the electronic device **101**. For example, the (1-2)th side wall portion **512** may operate as the (1-7)th antenna disposed in the (1-7)th antenna region **817** of FIG. **8**. According to an embodiment, when operating as the antenna, the (1-2)th side wall portion **512** may include a portion **1011** to which the feeding portion of the antenna is connected and a portion **1012** to which the switch portion of the antenna is connected. According to an embodiment, when the (1-2)th side wall portion **512** operates as the antenna, the first bracket **610** may include a portion **1013** to which the ground portion of the antenna is connected. According to an embodiment, the portion **1011** to which the feeding portion of the antenna included in the (1-2)th side wall portion **512** is connected, the portion **1012** to which the switch portion of the antenna is connected, and the portion **1013** to which the ground portion of the antenna included in the first bracket **610** is connected may be electrically connected to a first flexible printed circuit board (FPCB) **1110**, and the first flexible printed circuit board **1110** may be electrically connected to the first printed circuit board **710** in first housing structure **410** via a connector. For example, when the (1-2)th side wall portion **512** operates as the antenna, the ground portion of the antenna may be connected to the portion **1013** to which the ground portion of the antenna included in the first bracket **610** is connected, and may be indirectly electrically connected to the ground portion of the first printed circuit board **710** via the first flexible printed circuit board **1110**. According to another embodiment, when the (1-2)th side wall portion **512** operates as the antenna, the portion **1011** to which the feeding portion of the antenna is connected without the portion **1012** to which the switch portion of the antenna is connected and the antenna portion **1013** to which the ground portion of the

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antenna is connected may be included in the (1-2)th side wall portion **512** and the first bracket **610**, respectively.

Referring to FIG. **11**, the first side wall structure **510** and the first bracket **610** are made of the same metal material (e.g., Al), and the first side wall structure **510** and the first bracket **610** are integrally configured to minimize resistance due to the bonding structure of the first side structure **510** and the first bracket **610**, so that the portion **1013** to which the ground portion of the antenna is connected may be located in the first bracket **610**.

Referring to FIG. **12**, the second side wall structure **520** may include a (2-2)th side wall portion **522** separated from a (2-1)th side wall portion **521** and a (2-3)th side wall portion **523** by a (2-1)th split portion **541** and a (2-2)th split portion **542**. According to an embodiment, the (2-2)th side wall portion **522** may operate as an antenna of the electronic device **101**. According to an embodiment, when operating as the antenna, the (2-2)th side wall portion **522** may include a portion **1021** to which the feeding portion of the antenna is connected, a portion **1022** to which the switch of the antenna is connected, and a portion **1023** to which the ground portion of the antenna is connected. According to an embodiment, the portion **1021** to which the feeding portion of the antenna included in the (2-2)th side wall portion **522** is connected, the portion **1022** to which the switch portion of the antenna is connected, and the portion **1023** to which the ground portion of the antenna is connected may be electrically connected to a second flexible printed circuit board **1120**, and the second flexible printed circuit board **1120** may be electrically connected to the third printed circuit board **730** in second housing structure **420** via a connector. For example, when the (2-2)th side wall portion **522** operates as the antenna, the ground portion of the antenna may be connected to the portion **1023** to which the ground portion of the antenna included in the (2-2)th side wall portion **522** is connected, and may be directly electrically connected to the ground portion of the third printed circuit board **730** via the second flexible printed circuit board **1120**. According to another embodiment, when the (2-2)th side wall portion operates as the antenna, the portion **1021** to which the feeding portion of the antenna is connected without the portion **1022** to which the switch portion of the antenna is connected and the portion **1023** to which the ground portion of the antenna is connected may be included in the (2-2)th side wall portion **522**.

In FIG. **12**, when the second side wall structure **520** and the second bracket **620** are made of different metals (e.g., the second side wall structure **520** is made of Al and the second bracket **620** is made of Mg), it is possible to prevent an impedance mismatch or an increase in electrical resistance due to a difference in electrical conductivity and/or a difference in electrical conductivity between the different metals by placing the portion **1023** to which the ground of the antenna is connected on the second side wall structure **520** (e.g., the (2-2)th side wall portion **522**) together the portion **1022** to which the switch portion of the antenna is connected and the portion **1021** to which the feeding portion of the antenna is connected.

FIG. **13** is a cross-sectional view of an electronic device taken in a B-B' direction according to an embodiment of the disclosure.

Referring to FIG. **13**, the inside of the first side wall structure **510** of the electronic device **101** may include a recess **1210**. According to an embodiment, at least a portion of the first bracket **610** may be inserted into the recess **1210** to be electrically coupled to the first side wall structure **510**. According to an embodiment, at least a portion of the first

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printed circuit board **710** may be electrically coupled to the first bracket **610**, and another portion of the first printed circuit board **710** may be electrically coupled to the first side wall structure **510**. According to an embodiment, at least a portion of the rear cover **240** (e.g., the first rear cover **240** in FIGS. **2** to **4**) may be coupled to the first bracket **610**, and another portion of the rear cover **240** may be coupled to the first side wall structure **510**.

FIG. **14** is a cross-sectional view of an electronic device taken in a C-C' direction according to an embodiment of the disclosure. FIG. **15** is a cross-sectional view of an electronic device taken in the D-D' direction according to an embodiment of the disclosure. FIG. **14** illustrates a connective relationship between a second side wall structure **520** and a second bracket **620** when at least a portion of a second side wall structure **520** operates as an antenna. FIG. **15** illustrates a connective relationship between a second side wall structure **520** and a second bracket **620** when at least a portion of a second side wall structure **520** does not operate as an antenna.

Referring to FIG. **14**, at least a portion of the second side wall structure **520** of the electronic device **101** may be bonded by an adhesive member **1310** to at least a portion of a plate **232** (e.g., the plate **232** in FIG. **4**) on which a display panel **231** (e.g., the display panel **231** in FIG. **4**) is seated and which is disposed between the display panel **231** and the second bracket **620**, and may be bonded to at least a portion of the rear cover **250** (e.g., the second rear cover **250** in FIGS. **2** to **4**) by an adhesive member **1310**. According to an embodiment, the second bracket **620** may be separated from the second bezel structure **520** through a space D therebetween. According to an embodiment, the space D may be filled with a non-conductive material to physically interconnect the second bracket **620** and the second bezel structure **520**.

Referring to FIG. **15**, the inside of the second side wall structure **520** of the electronic device **101** may include a recess **1410**. According to an embodiment, at least a portion of the second bracket **620** may be inserted into the recess **1410** to be electrically coupled to the second side wall structure **520**. According to an embodiment, at least a portion of the third printed circuit board **730** may be electrically coupled to the second bracket **620**, and another portion of the third printed circuit board **730** may be electrically coupled to the second side wall structure **520**. According to an embodiment, at least a portion of the rear cover **250** (e.g., the second rear cover **250** in FIGS. **2** to **4**) may be coupled to the second bracket **620**, and another portion of the rear cover **250** may be coupled to the second side wall structure **520**.

FIG. **16** is a view illustrating an embodiment in which a second housing structure of an electronic device of FIG. **5** is formed of a same metal material according to an embodiment of the disclosure. FIG. **17** is a cross-sectional view of an electronic device of FIG. **16** taken in a E-E' direction according to an embodiment of the disclosure.

Referring to FIGS. **16** and **17**, in the electronic device **101**, a portion of the second side wall structure **520** included in the second housing structure **420** may include a non-conductive material (e.g., ceramic) for aesthetic appearance. According to an embodiment, the second side wall structure **520** included in the second housing structure **420** of the electronic device **101** may include a (2-1)th side wall structure **520-1** including a non-conductive material and a (2-2)th side wall structure **520-2** including a metal material. According to an embodiment, the second housing structure **420** may include a structure in which at least a portion of the

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(2-2)th side wall structure **520-2** is separated from the second bracket **620**. According to an embodiment, between the at least one portion of the (2-2)th side wall structure **520-2** and the second bracket **620**, a non-conductive material **1510** may be injection molded in order to physically interconnect the at least a portion of the (2-2)th side wall structure **520-2** and a second bracket **620**.

According to an embodiment, the (2-2)th side wall structure **520-2** may include at least one pair of split portions **1610**. According to an embodiment, the at least one pair of split portions **1610** may be formed of an insulator. According to an embodiment, the at least one pair of split portions **1610** may be molded into the (2-2)th side wall structure through a double-injection molding method or an insert molding method of a synthetic resin. However, the disclosure is not limited thereto, and the at least one pair of split portions **1610** may be made of various materials having an insulating property. According to an embodiment, the (2-2)th side wall structure **520-2** may include a side wall portion **1710** split by the one pair of split portions **1610**.

According to an embodiment, the side wall portion **1710** may operate as an antenna of the electronic device **101**. According to an embodiment, the side wall portion **1710** may include a first portion **1710-1** that extends to the (2-1)th side wall structure **520-1** to emit a signal, and a second portion **1710-2** that includes at least one antenna element **1711** (e.g., a switch portion of the antenna, a feeding portion of the antenna, or a ground portion of the antenna). According to an embodiment, the at least one antenna element **1711** may be electrically connected to the third printed circuit board **730** in the second housing structure **420** via an antenna connection portion **1721** (e.g., a C-clip).

According to an embodiment, when a portion of the at least one antenna element **1711** included in the first portion **1710-1** of the side wall portion **1710** is located on the second bracket **620**, the (2-2)th side wall structure **520-2** including the side wall portion **1710** and the second bracket **620** may include the same metal material in order to prevent the occurrence of signal distortion due to a difference in electrical conductivity between different metals and/or an increase in impedance mismatch or electrical resistance due to the difference in electrical conductivity.

According to an embodiment, the second housing structure **420** may include a display **230** (e.g., the display **230** in FIGS. **2** to **4**) and a window glass **233** that protects the display **230**. According to an embodiment, the second housing structure **420** may include a sub-display **252** (e.g., the sub-display **252** in FIGS. **2** to **4**) disposed in a region of the rear cover **250** (e.g., the second rear cover **250** in FIGS. **2** to **4**).

According to an embodiment, the second housing structure **420** may be relatively heavy in weight compared to the first housing structure (not illustrated) due to an additional configuration such as the sub-display **252**. Thus, in order to reduce the weight of the second housing structure **420**, the (2-2)th side wall structure **520-2** and the second bracket **620** included in the second housing structure **420** may include a metal material having a low specific gravity. For example, the (2-2)th side wall structure **520-2** and the second bracket **620** may include a Mg material.

According to an embodiment, the (2-2)th side wall structure **520-2** and the second bracket **620** included in the second housing structure **420** may include a metal material having a low specific gravity (e.g., Mg) for weight reduction. According to an embodiment, since the side wall portion **1710** included in the (2-2)th side wall structure **520-2** includes the same metal material as the second bracket **620**,

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signal distortion due to an increase in impedance mismatch or electrical resistance due to a difference in electrical conductivity may not occur even when a portion of the at least one antenna element 1711 included in the first portion 1710-1 of the side wall portion 1710 is located in the second bracket 620. According to an embodiment, since the (2-2)th side wall structure 520-2 and the second bracket 620 included in the second housing structure 420 may include the same metal having low electrical conductivity (e.g., Mg), the second housing structure 420 may further include a signal amplifier (e.g., a power amplifier module (PAM)) in order to compensate for signal distortion (e.g., signal loss) due to low electrical conductivity.

According to an embodiment of the disclosure, an electronic device (e.g., 101 in FIGS. 1 to 5) may include a foldable housing including: a hinge structure (e.g., the hinge structure 264 in FIG. 4), a first housing (e.g., the first housing 210 in FIGS. 2 to 4 or the first housing 410 in FIGS. 5 and 6) connected to the hinge structure and including a first side wall structure and a first bracket located inside the first side wall structure, and a second housing (e.g., the second housing 220 in FIGS. 2 to 4 or the second housing 420 in FIGS. 5 and 6) connected to the hinge structure and including a second side wall structure and a second bracket located inside the second side wall structure, the second housing being folded with the first housing about the hinge structure, a display panel (e.g., the display panel 231 in FIGS. 2 and 3) extending from the first housing to the second housing across the hinge structure and configured to output a screen, and a second antenna structure (e.g., the second antenna structure 542 in FIGS. 5 and 6) including a second ground and a second feeding portion disposed on the second side wall structure of the second housing, wherein the first side wall structure of the first housing and the second side wall structure of the second housing may include a first metal material, and the first bracket of the first housing includes the first metal material, and the second bracket of the second housing may include a second metal material.

In the electronic device (e.g., 101 of FIGS. 1 to 5) according to an embodiment, the first metal material has higher electrical conductivity and thermal conductivity than the second metal material, and the second metal material may have a lower specific gravity than the first metal.

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment, the first metal material may be aluminum (Al).

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment, the second metal material may be a magnesium (Mg) material.

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment, the second antenna structure may further include a second antenna switch disposed on the second side wall structure of the second housing.

The electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment may further include a first antenna structure provided by a portion of the first housing, wherein the first antenna structure may include a first feeding portion disposed on the first side wall structure, and a first ground disposed on the first bracket.

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment, the first antenna structure may further include a first antenna switch disposed on the first side wall structure of the first housing.

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment, the first side wall structure and the first bracket may be integrally configured.

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In the electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment, the second side wall structure and the second bracket may be separably configured.

The electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment may further include a sub-display panel that is disposed on a surface of the second housing that is opposite to the surface on which the display panel is disposed.

The electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment may further include a first printed circuit board supported by the first bracket, and a second printed circuit board supported by the second bracket.

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment, the first printed circuit board may have a structure in which one or more heating elements, which are heat sources, are disposed in multiple layers.

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment, the second printed circuit board may have a structure in which one or more heating elements, which are heat sources, are disposed in a single layer.

The electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment may further include a heat dissipation structure disposed on the first printed circuit board.

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to an embodiment, the heat dissipation structure may include: a shield can mounted on one surface of the first printed circuit board in a state of accommodating the heating elements and having a shape surrounding at least some of the heating elements, a heat transfer member disposed on the heating elements and configured to transfer heat generated by the heating elements, and a heat diffusion structure disposed on at least a portion of the heat transfer member and the shield can to provide a function of transferring outward heat generated by the heating elements, wherein the heat diffusion structure may be disposed to be at least partially in contact with the first bracket.

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to various embodiments, the heat generated from the heating element may move to the first bracket via the heat transfer member and the heat diffusion structure.

In the electronic device (e.g., 101 in FIGS. 1 to 5) according to various embodiments, the heat dissipation structure may further include a vapor chamber that is disposed on the first bracket and absorbs or diffuses heat transferred thereto from the heat diffusion structure via the first bracket.

In an electronic device (e.g., 101 in FIGS. 1 to 5) according to various embodiments, the heat diffusion structure may include a heat dissipation member and an elastic member disposed along an edge of the heat dissipation member, the heat dissipation member may be disposed to be in direct contact with the heat transfer member and is configured to diffuse heat transferred thereto from the heating element, and the elastic member may be connected to the heat dissipation member and disposed on a portion of the shield can.

In the electronic device (e.g., 101 of FIGS. 1 to 5) according to various embodiments, the second side wall structure may be coupled, by an adhesive member, to at least a portion of a plate on which the display panel is seated and disposed between the display panel and the second bracket, and a separated space may be included between the second side wall structure and the second bracket.

The electronic device (e.g., 101 in FIGS. 1 to 5), may further include a second recess provided inside the second side wall structure, wherein the second bracket may be at

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least partially inserted into the second recess to be coupled to the second side wall structure.

While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

What is claimed is:

1. An electronic device comprising:
a foldable housing including:
a hinge structure,
a first housing connected to the hinge structure, and including a first side wall structure and a first bracket supporting at least a first electrical component of the electronic device, wherein at least a part of the first bracket is integrally coupled to at least a part of the first side wall structure, and
a second housing connected to the hinge structure, and including a second side wall structure and a second bracket supporting at least a second electrical component of the electronic device, wherein at least a part of the second bracket is integrally coupled to at least a part of the second side wall structure, and wherein the second housing is foldable with the first housing about the hinge structure;
a display panel extending from the first housing to the second housing across the hinge structure and configured to output a screen; and
a first antenna structure including at least a portion of the second side wall structure that is electrically connected directly to a first ground portion and electrically connected directly to a first feeding portion,
wherein the first side wall structure, the first bracket, and the second side wall structure include a first metal material,
wherein the second bracket includes a second metal material, and
wherein the first metal material and second metal material are different metal materials having different properties.
2. The electronic device of claim 1, wherein the first metal material has higher electrical conductivity and specific gravity than the second metal material.
3. The electronic device of claim 2,
wherein the first metal material is an aluminum (Al) material, and
wherein the second metal material is a magnesium (Mg) material.
4. The electronic device of claim 1, wherein the first metal material has higher thermal conductivity than the second metal material.
5. The electronic device of claim 1, wherein the at least the portion of the second side wall structure is electrically connected directly to a first antenna switch.
6. The electronic device of claim 1, further comprising:
a second antenna structure including at least a portion of the first side wall structure that is electrically connected directly to a second feeding portion,
wherein that at least the portion of the first side wall structure is one of electrically connected directly to a second ground portion or electrically connected indirectly to the second ground portion via the first bracket.
7. The electronic device of claim 6, wherein the at least the portion of the first side wall structure is electrically connected directly to a second antenna switch.

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8. The electronic device of claim 1, further comprising:
a sub-display panel disposed in the second housing, wherein the sub-display panel and the display panel are disposed on opposite surfaces of the second housing.

9. The electronic device of claim 1,
wherein the first electrical component includes a first printed circuit board, and
wherein the second electrical component includes a second printed circuit board.

10. The electronic device of claim 9, wherein the first printed circuit board has a structure in which one or more heating elements, which are heat sources for generating heat, are arranged in multiple layers.

11. The electronic device of claim 9, wherein the second printed circuit board has a structure in which one or more heating elements, which are heat sources for generating heat, are arranged in a single layer.

12. The electronic device of claim 9, further comprising:
a heat dissipation structure disposed on the first printed circuit board.

13. The electronic device of claim 12,
wherein the heat dissipation structure includes:

a shield can mounted on one surface of the first printed circuit board in a state of accommodating heating elements and having a shape surrounding at least some of the heating elements;

a heat transfer member disposed on the heating elements and configured to transfer heat generated by the heating elements; and

a heat diffusion structure disposed on at least a portion of the heat transfer member and the shield can to provide a function of transferring outward heat generated by the heating elements, and

wherein the heat diffusion structure is disposed to be at least partially in contact with the first bracket.

14. The electronic device of claim 13, wherein the heat generated by the heating elements moves to the first bracket via the heat transfer member and the heat diffusion structure.

15. The electronic device of claim 13, wherein the heat dissipation structure further includes a vapor chamber disposed on the first bracket to absorb or diffuse heat transferred thereto from the heat diffusion structure via the first bracket.

16. The electronic device of claim 13,

wherein the heat diffusion structure includes a heat dissipation member and an elastic member disposed along an edge of the heat dissipation member,

wherein the heat dissipation member is disposed to be in direct contact with the heat transfer member and is configured to diffuse heat transferred thereto from the heating elements, and

wherein the elastic member is connected to the heat dissipation member and disposed on a portion of the shield can.

17. The electronic device of claim 1, wherein the second side wall structure is coupled, by an adhesive member, to at least a portion of a plate on which the display panel is seated and disposed between the display panel and the second bracket.

18. The electronic device of claim 1,

wherein a recess is formed into the at least the part of the second side wall structure, and

wherein the at least the part of the second bracket that is integrally coupled to the at least the part of the second side wall structure is at least partially disposed in the recess.